



DC – AI – Routing – Switching – Security – Wireless – Automation

SRX EVPN/VXLAN T5 oIPSEC

An HPE Juniper Networking Techpost

Karel Hendrych

2024-05-27

HPE



Contents

1	SRX EVPN/VXLAN T5 oIPSEC	3
1.1	Introduction	3
1.2	The Demo Setup	3
1.3	Brief Look at EVPN/VXLAN Type 5 Settings	4
1.4	Tenant-1 (blue) intra-VRF Traffic	5
1.5	Tenant-1 (blue) to Tenant-2 (green)	7
1.6	Tenant-1 (blue) to Tenant-3 (red) – Overlapping IP Prefixes	11
1.7	MTU Considerations	13
1.8	Appendix 1 – SRX Transit VXLAN Tunnel Inspection	14
1.9	Appendix 2 – Linux Network Namespace Sample Configs	16
1.10	Appendix 3 – SRX Configuration Files	17
1.11	Useful links	25
1.12	Glossary	25
1.13	Acknowledgements	26

List of Figures

1	demo topology	3
2	blue tenant VRF traffic	5
3	blue tenant VRF to green tenant VRF (vSRX-right handled)	8
4	vSRX right interlinking tenant VRFs using common-1 instance	8
5	U-turn on vSRX-right blue - green tenant VRF	10
6	Overlapping prefix handling on vSRX-right for blue/red tenant VRFs	11
7	Details of vSRX-right NAT approach	12
8	vSRX-r1 inspection and enforcement of unencrypted VXLAN	15

1 SRX EVPN/VXLAN T5 oIPSEC

Karel Hendrych - 05/27/2024

A practical yet simple demonstration of the SRX EVPN/VXLAN Type 5 ip-prefix-routes feature and related firewall policy processing across multiple tenants, including an example of communication between overlapping IP prefixes. By utilizing an IPSEC underlay in an otherwise data center-centric feature set, this TechPost article effectively demonstrates the potential expansion into WAN scenarios.

1.1 Introduction

Prior to Junos 22.4R1 [EVPN/VXLAN Type 5 support](#), SRX had a phased roll-out in Junos 20.4-21.1 of the feature set for [L4-L7 inspection of transit VXLAN tunnels](#), as briefly demonstrated in Appendix 1. However, things became more interesting with 22.4R1 due to the ability to significantly reduce the number of BGP peerings when connecting to EVPN fabric in a role of intra/inter VRF segmentation firewall, where VXLAN VNI is mapped into VRF. Type 5 is very efficient because it sends subnets in BGP advertisements rather than MAC/IP ones as with Type 2.

In a traditional design without EVPN/VXLAN, one BGP peering per interface belonging to a routing instance is required, potentially creating scalability challenges. However, everything comes at a price, and in this case, there is slightly more configuration required initially compared to a classic L3 firewall. Nevertheless, adding subsequent tenants in an EVPN/VXLAN SRX setup dramatically reduces configuration overhead of the classic approach.

Previously, without additional encapsulations like MPLSoGRE, achieving multi-tenancy using IPSEC was not straightforward, requiring various tricks with multiple IKE/CHILD security associations originating from alias addresses. With VXLAN host-to-host (effectively loopback-to-loopback overlay) communication, securing transit by folding all tenant traffic into IPSEC becomes easy. Type 5 support is entirely handled in SRX Flow mode, eliminating the need for selective packet services as seen in MPLSoGRE designs.

1.2 The Demo Setup

Let's consider the following setup – basically two vSRX instances, each with three EVPN/VXLAN T5 stretched tenant VRFs – blue, green, red - connected by vSRX based underlay router.

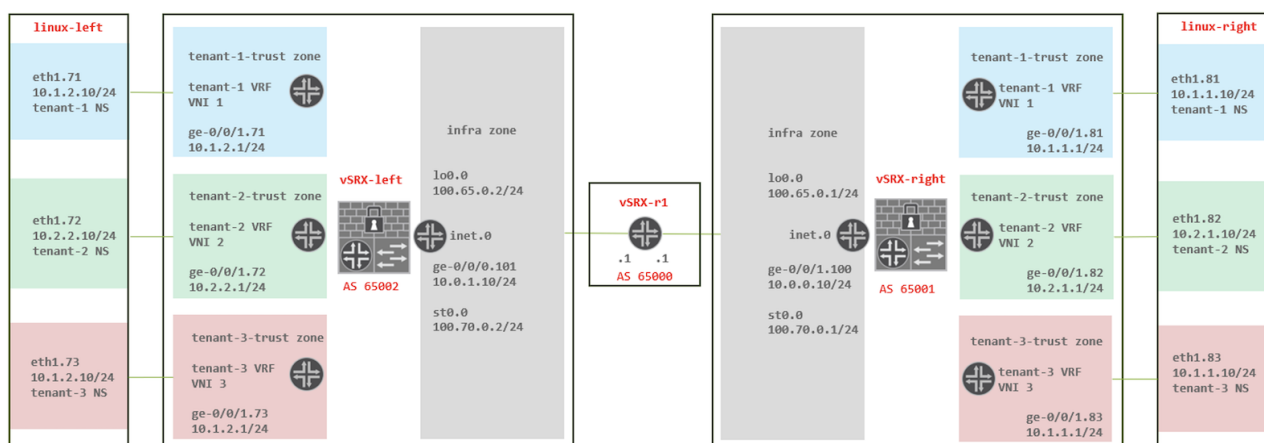


Figure 1: demo topology

Demo setup features:

- vSRX-left acts more like a spoke, permitting any traffic in/out of individual VRFs and local interfaces for individual tenants, with no route leaking between VRFs.
- vSRX-right, in a sort of hub role, filters traffic between tenants. For example, traffic from left blue (VNI 1) to left green (VNI 2) is enforced on vSRX-right, effectively causing traffic to always U-turn on vSRX-right.
- Tenant-1 can reach tenant-2 and tenant-3 on specific protocols, but not the other way around.
- Tenant-3 has overlapping IP address space with tenant-1. The communication between them is facilitated on vSRX-right using static NAT.
- Communication between loopbacks sourcing VXLAN traffic is encrypted using route-based IPSEC VPN, with one CHILD SA handling all loopback-loopback VXLAN traffic.
- Decapsulated traffic originates in the infra zone, the zone where the underlay interfaces are bound to.
- For convenience, two Linux endpoints simulating all the tenants are using network namespaces.
- By disabling IPSEC, traffic shifts to clear-text underlay without any loss.
- Non-encrypted VXLAN communication is inspected and enforced on vSRX-r1

1.3 Brief Look at EVPN/VXLAN Type 5 Settings

In the example below, vSRX-left is configured with two BGP underlay peerings, vSRX-r1 and towards vSRX-right IPSEC tunnel, where to the local lo0.0 IP is exported. vSRX-right takes precedence when the IPSEC VPN is up. A new part of the SRX configuration involves EVPN signaling, establishing a loopback-to-loopback overlay peering with vSRX-right. > Note - complete configuration files are present in Appendix 3, in the latest part of this article.

```

1 set protocols bgp group underlay bfd-liveness-detection minimum-interval 500
2 set protocols bgp group underlay bfd-liveness-detection multiplier 3
3 set protocols bgp group underlay neighbor 10.0.1.1 description r1
4 set protocols bgp group underlay neighbor 10.0.1.1 export export-underlay
5 set protocols bgp group underlay neighbor 10.0.1.1 peer-as 65000
6 set protocols bgp group underlay neighbor 10.0.1.1 local-as 65002
7 set protocols bgp group underlay neighbor 100.70.0.1 description vsrx-right-st0
8 set protocols bgp group underlay neighbor 100.70.0.1 export export-underlay
9 set protocols bgp group underlay neighbor 100.70.0.1 peer-as 65001
10 set protocols bgp group underlay neighbor 100.70.0.1 local-as 65002
11
12 set protocols bgp group overlay multihop
13 set protocols bgp group overlay local-address 100.65.0.2
14 set protocols bgp group overlay family evpn signaling
15 set protocols bgp group overlay bfd-liveness-detection minimum-interval 500
16 set protocols bgp group overlay bfd-liveness-detection multiplier 5
17 set protocols bgp group overlay neighbor 100.65.0.1 peer-as 65101
18 set protocols bgp group overlay neighbor 100.65.0.1 local-as 65102

```

Next part is configuration of individual tenant VRFs, where:

- ip-prefix-routes knobs
 - enable Type 5 support
 - define VXLAN encapsulation
 - assign unique VNI
 - refer to export policy
- local interface gets bound to VRF
- route distinguisher is ensuring uniqueness of routes in case of overlaps
- vrf-target is for purpose of automatic import into matching VRFs across the topology

```

1 set routing-instances tenant-1 instance-type vrf
2 set routing-instances tenant-1 protocols evpn ip-prefix-routes advertise direct-nexthop
3 set routing-instances tenant-1 protocols evpn ip-prefix-routes encapsulation vxlan
4 set routing-instances tenant-1 protocols evpn ip-prefix-routes vni 1
5 set routing-instances tenant-1 protocols evpn ip-prefix-routes export export-tenant-1
6 set routing-instances tenant-1 interface ge-0/0/1.71
7 set routing-instances tenant-1 route-distinguisher 100.65.0.1:1
8 set routing-instances tenant-1 vrf-target target:65001:1
9 set policy-options policy-statement export-tenant-1 term 1 from interface ge-0/0/1.71
10 set policy-options policy-statement export-tenant-1 term 1 then accept
11 set policy-options policy-statement export-tenant-1 term 100 then reject
12
13 set routing-instances tenant-2 instance-type vrf
14 set routing-instances tenant-2 protocols evpn ip-prefix-routes advertise direct-nexthop
15 set routing-instances tenant-2 protocols evpn ip-prefix-routes encapsulation vxlan
16 set routing-instances tenant-2 protocols evpn ip-prefix-routes vni 2
17 set routing-instances tenant-2 protocols evpn ip-prefix-routes export export-tenant-2
18 set routing-instances tenant-2 interface ge-0/0/1.72
19 set routing-instances tenant-2 route-distinguisher 100.65.0.1:2
20 set routing-instances tenant-2 vrf-target target:65001:2
21 set policy-options policy-statement export-tenant-2 term 1 from interface ge-0/0/1.72
22 set policy-options policy-statement export-tenant-2 term 1 then accept
23 set policy-options policy-statement export-tenant-2 term 100 then reject

```

1.4 Tenant-1 (blue) intra-VRF Traffic

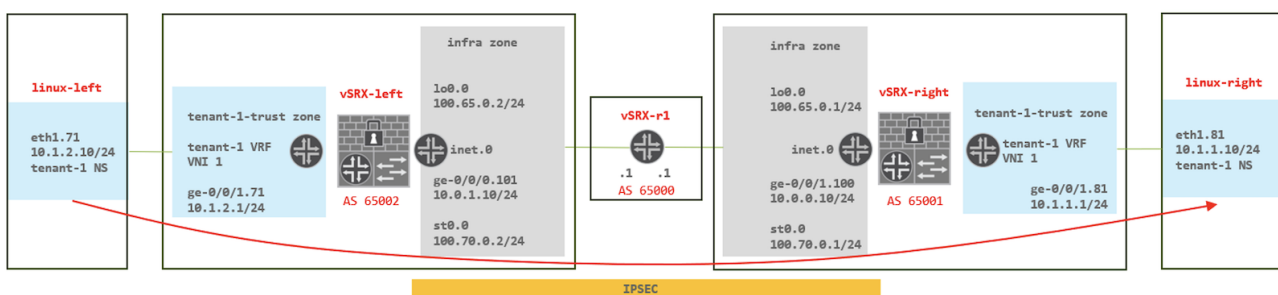


Figure 2: blue tenant VRF traffic

vSRX-left tenant-1 is receiving default gateway from vSRX-right and exports local interface prefix.

```

1 vsrx-left> show evpn ip-prefix-database l3-context tenant-1
2 L3 context: tenant-1
3
4 IPv4->EVPN Exported Prefixes
5 Prefix EVPN route status
6 10.1.2.0/24 Created
7
8 EVPN->IPv4 Imported Prefixes
9 Prefix Etag
10 0.0.0.0/0 0
11 Route distinguisher VNI/Label Router MAC Nexthop/Overlay GW/ESI
12 100.65.0.1:1 1 4c:96:14:a8:1b:b0 100.65.0.1

```

Correspondingly vSRX-right is exporting default gateway and imports prefix of locally connected network to vSRX-left, for example:

```

1 vsrx-right> show evpn ip-prefix-database l3-context tenant-1
2 L3 context: tenant-1
3
4 IPv4->EVPN Exported Prefixes
5 Prefix EVPN route status
6 0.0.0.0/0 Created
7
8 EVPN->IPv4 Imported Prefixes
9 Prefix Etag
10 10.1.2.0/24 0
11 Route distinguisher VNI/Label Router MAC Nexthop/Overlay GW/ESI

```

```
12 100.65.0.1:1 1 4c:96:14:2e:cf:b0 100.65.0.2
```

For completeness BGP, routing information from vSRX-1, including EVPN control plane – underlay is preferred via IPSEC tunnel interface st0.0, tenant-2 and tenant-3 have similar layout as tenant-1

```
1 vsrx-left> show route protocol bgp
2
3 inet.0: 10 destinations, 11 routes (10 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 10.0.0.0/24 *[BGP/170] 1d 07:57:12, localpref 100
7   AS path: 65000 I, validation-state: unverified
8   > to 10.0.1.1 via ge-0/0/0.101
9 100.65.0.1/32 *[BGP/170] 1d 00:33:39, localpref 100
10  AS path: 65001 I, validation-state: unverified
11  > to 100.70.0.1 via st0.0
12  [BGP/170] 1d 07:57:12, localpref 100
13  AS path: 65000 65001 I, validation-state: unverified
14  > to 10.0.1.1 via ge-0/0/0.101
15
16 bgp.evpn.0: 6 destinations, 6 routes (6 active, 0 holddown, 0 hidden)
17 + = Active Route, - = Last Active, * = Both
18
19 5:100.65.0.1:1::0::0.0.0.0::0/248
20   *[BGP/170] 1d 00:44:58, localpref 100, from 100.65.0.1
21   AS path: 65101 I, validation-state: unverified
22   > to 100.70.0.1 via st0.0
23 5:100.65.0.1:2::0::0.0.0.0::0/248
24   *[BGP/170] 1d 00:44:58, localpref 100, from 100.65.0.1
25   AS path: 65101 I, validation-state: unverified
26   > to 100.70.0.1 via st0.0
27 5:100.65.0.1:3::0::0.0.0.0::0/248
28   *[BGP/170] 1d 00:44:58, localpref 100, from 100.65.0.1
29   AS path: 65101 I, validation-state: unverified
30   > to 100.70.0.1 via st0.0
31
32 tenant-1.evpn.0: 2 destinations, 2 routes (2 active, 0 holddown, 0 hidden)
33 + = Active Route, - = Last Active, * = Both
34
35 5:100.65.0.1:1::0::0.0.0.0::0/248
36   *[BGP/170] 1d 00:44:58, localpref 100, from 100.65.0.1
37   AS path: 65101 I, validation-state: unverified
38   > to 100.70.0.1 via st0.0
39
40 tenant-2.evpn.0: 2 destinations, 2 routes (2 active, 0 holddown, 0 hidden)
41 + = Active Route, - = Last Active, * = Both
42
43 5:100.65.0.1:2::0::0.0.0.0::0/248
44   *[BGP/170] 1d 00:44:58, localpref 100, from 100.65.0.1
45   AS path: 65101 I, validation-state: unverified
46   > to 100.70.0.1 via st0.0
47
48 tenant-3.evpn.0: 2 destinations, 2 routes (2 active, 0 holddown, 0 hidden)
49 + = Active Route, - = Last Active, * = Both
50
51 5:100.65.0.1:3::0::0.0.0.0::0/248
52   *[BGP/170] 1d 00:44:58, localpref 100, from 100.65.0.1
53   AS path: 65101 I, validation-state: unverified
54   > to 100.70.0.1 via st0.0
```

Permissive security policies within tenant-1 – notice the use of Destination and Source VRF group match for traffic sourced or destined to the infra zone binding underlay interface. That's how in the current design the source and destination zone are determined with an option to narrow down using VRF matching. The firewall policy on vSRX-right is the same.

```
1 [edit security policies from-zone tenant-1-trust to-zone infra]
2 vsrx-left# show | display set relative | display inheritance no-comments
3 set policy tenant-1 match source-address any
4 set policy tenant-1 match destination-address any
5 set policy tenant-1 match application any
6 set policy tenant-1 match destination-l3vpn-vrf-group tenant-1
7 set policy tenant-1 then permit
8 set policy tenant-1 then log session-close
9
10 [edit security policies from-zone infra to-zone tenant-1-trust]
```

```

11 vsrx-left# show | display set relative | display inheritance no-comments
12 set policy tenant-1 match source-address any
13 set policy tenant-1 match destination-address any
14 set policy tenant-1 match application any
15 set policy tenant-1 match source-l3vpn-vrf-group tenant-1
16 set policy tenant-1 then permit
17 set policy tenant-1 then log session-close

```

Finally, this is Linux-left establishing an SSH connection to Linux-right in the tenant-1 network namespace and related session listing from both the left and right vSRX instances. Sample Linux network namespace configuration for easy lab work (avoiding the need to have many endpoints) is described in Appendix 2.

```

1 linux-left:~# alias t1
2 alias t1='ip netns exec tenant-1'
3
4 linux-left:~# t1 ssh 10.1.1.10
5 Linux linux-right 5.10.0-29-amd64 #1 SMP Debian 5.10.216-1 (2024-05-03) x86_64
6 linux-right:~#
7
8 vsrx-left> show security flow session destination-port 22
9 Session ID: 1414, Policy name: tenant-1/4, Timeout: 1798, Session State: Valid
10 In: 10.1.2.10/49832 --> 10.1.1.10/22;tcp, If: ge-0/0/1.71, Pkts: 22, Bytes: 3928,
11 Out: 10.1.1.10/22 --> 10.1.2.10/49832;tcp, L3VPN VRF Group: tenant-1, If: st0.0, Pkts: 23,
    Bytes: 4952
12 Total sessions: 1
13
14 vsrx-right> show security flow session destination-port 22
15 Session ID: 1401, Policy name: tenant-1/6, Timeout: 1792, Session State: Valid
16 In: 10.1.2.10/49832 --> 10.1.1.10/22;tcp, L3VPN VRF Group: tenant-1, If: st0.0, Pkts: 22,
    Bytes: 3928,
17 Out: 10.1.1.10/22 --> 10.1.2.10/49832;tcp, If: ge-0/0/2.81, Pkts: 23, Bytes: 4952,
18 Total sessions: 1

```

Corresponding close log record from vSRX-right:

```

1 RT_FLOW - RT_FLOW_SESSION_CLOSE [junos@2636.1.1.1.2.129 reason="TCP FIN" source-address
    ="10.1.2.10" source-port="49832" destination-address="10.1.1.10" destination-port="22"
    connection-tag="0" service-name="junos-ssh" nat-source-address="10.1.2.10" nat-source-port
    ="49832" nat-destination-address="10.1.1.10" nat-destination-port="22" nat-connection-tag
    ="0" src-nat-rule-type="N/A" src-nat-rule-name="N/A" dst-nat-rule-type="N/A" dst-nat-rule-
    name="N/A" protocol-id="6" policy-name="tenant-1" source-zone-name="infra" destination-zone-
    name="tenant-1-trust" session-id="1401" packets-from-client="169" bytes-from-client="16708"
    packets-from-server="305" bytes-from-server="28408" elapsed-time="16953" application="
    UNKNOWN" nested-application="UNKNOWN" username="N/A" roles="N/A" packet-incoming-interface="
    st0.0" encrypted="UNKNOWN" application-category="N/A" application-sub-category="N/A"
    application-risk="-1" application-characteristics="N/A" secure-web-proxy-session-type="NA"
    peer-session-id="0" peer-source-address="0.0.0.0" peer-source-port="0" peer-destination-
    address="0.0.0.0" peer-destination-port="0" hostname="NA NA" src-vrf-grp="tenant-1" dst-vrf-
    grp="N/A" tunnel-inspection="Off" tunnel-inspection-policy-set="root" session-flag="0"
    source-tenant="N/A" destination-service="N/A" user-type="N/A"]

```

Note: to reduce log verbosity, a custom log format with selected columns could be used.

1.5 Tenant-1 (blue) to Tenant-2 (green)

Next example is tenant-1 connecting to tenant-2 VRF where vSRX-right in a hub role is exchanging routing information between routing instances and enforces firewall policies.

The first example is Linux-left in the tenant-1-trust zone connecting to Linux-right in the tenant-2-trust zone.

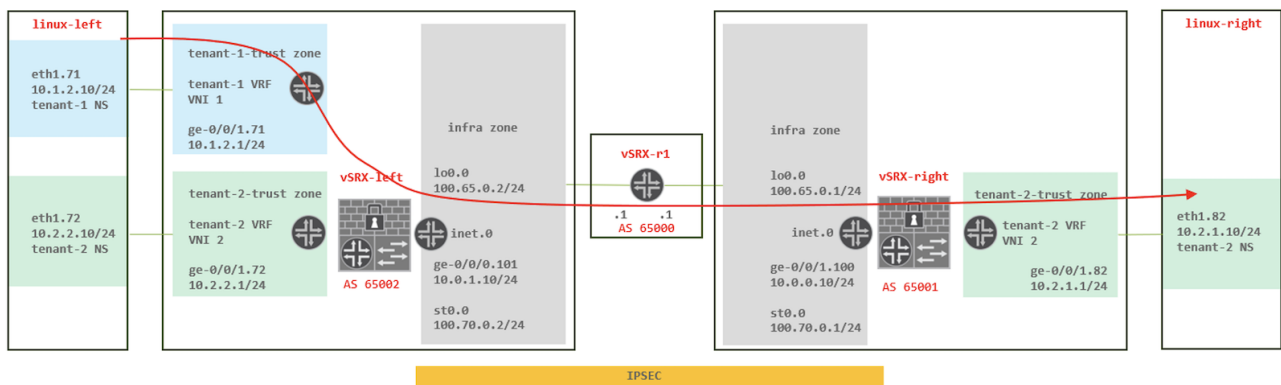


Figure 3: blue tenant VRF to green tenant VRF (vSRX-right handled)

Inter VRF routing is achieved using a shared instance named common-1 on vSRX-right, where the default routes from tenant-1 and tenant-2 VRFs are pointing. The use of the common instance addresses potential m:n route scalability issues if routes were full-mesh imported/exported directly. The layout of tenant and common VR route propagation:

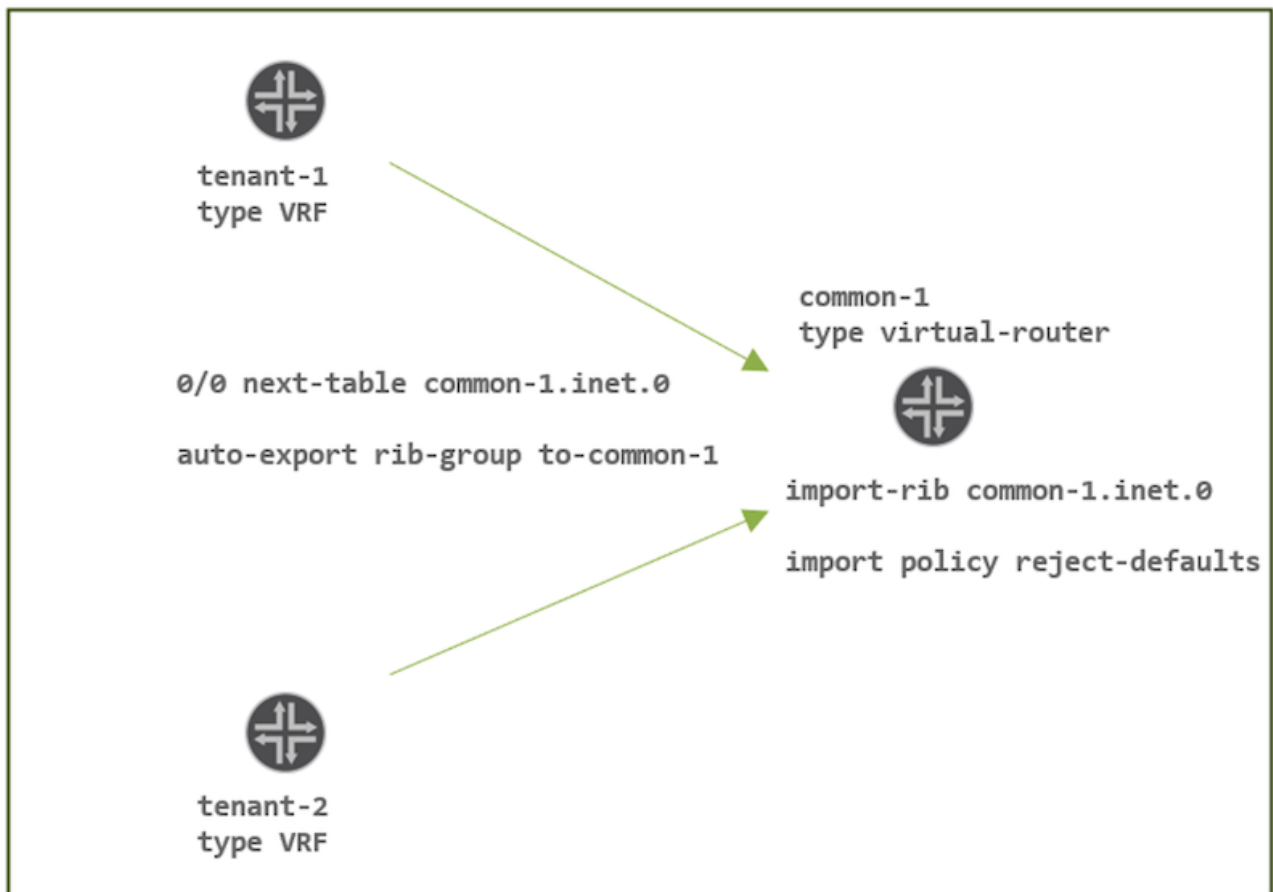


Figure 4: vSRX right interlinking tenant VRFs using common-1 instance

Routing table:

```
1 common-1.inet.0: 6 destinations, 6 routes (6 active, 0 holddown, 0 hidden)
2 + = Active Route, - = Last Active, * = Both
3
4 10.1.1.0/24 *[Direct/0] 1d 21:24:54
5 > via ge-0/0/2.81
```

```

6 10.1.1.1/32 *[Local/0] 1d 21:24:54
7     Local via ge-0/0/2.81
8 10.1.2.0/24 *[EVPN/170] 1d 14:12:22
9     > to 100.70.0.2 via st0.0
10 10.2.1.0/24 *[Direct/0] 1d 21:24:54
11     > via ge-0/0/2.82
12 10.2.1.1/32 *[Local/0] 1d 21:24:54
13     Local via ge-0/0/2.82
14 10.2.2.0/24 *[EVPN/170] 1d 14:12:22
15     > to 100.70.0.2 via st0.0
16
17 tenant-1.inet.0: 4 destinations, 4 routes (4 active, 0 holddown, 0 hidden)
18 + = Active Route, - = Last Active, * = Both
19
20 0.0.0.0/0 *[Static/5] 1d 21:25:21
21     to table common-1.inet.0
22 10.1.1.0/24 *[Direct/0] 1d 21:24:54
23     > via ge-0/0/2.81
24 10.1.1.1/32 *[Local/0] 1d 21:24:54
25     Local via ge-0/0/2.81
26 10.1.2.0/24 *[EVPN/170] 1d 14:12:22
27     > to 100.70.0.2 via st0.0
28
29 tenant-2.inet.0: 4 destinations, 4 routes (4 active, 0 holddown, 0 hidden)
30 + = Active Route, - = Last Active, * = Both
31
32 0.0.0.0/0 *[Static/5] 1d 21:25:21
33     to table common-1.inet.0
34 10.2.1.0/24 *[Direct/0] 1d 21:24:54
35     > via ge-0/0/2.82
36 10.2.1.1/32 *[Local/0] 1d 21:24:54
37     Local via ge-0/0/2.82
38 10.2.2.0/24 *[EVPN/170] 1d 14:12:22

```

Security policies permit intra-tenant traffic for tenant-2 like for tenant-1 in the previous example (1st policy below); however, tenant-1 can reach the local tenant-2 zone only using SSH (defined as a dynamic application based on Application Identification) and ICMP echo-request (2nd policy).

```

1 [edit security policies from-zone infra to-zone tenant-2-trust]
2 vsrx-right# show | display set relative
3 set policy tenant-2 match source-address any
4 set policy tenant-2 match destination-address any
5 set policy tenant-2 match application any
6 set policy tenant-2 match source-l3vpn-vrf-group tenant-2
7 set policy tenant-2 then permit
8 set policy tenant-2 then log session-close
9
10 set policy tenant-1 match source-address any
11 set policy tenant-1 match destination-address any
12 set policy tenant-1 match application any
13 set policy tenant-1 match dynamic-application junos:SSH
14 set policy tenant-1 match dynamic-application junos:ICMP-ECHO
15 set policy tenant-1 match source-l3vpn-vrf-group tenant-1
16 set policy tenant-1 then permit
17 set policy tenant-1 then log session-close

```

Session listing on vSRX-right – tenant-1 on Linux-left connecting to tenant-2 endpoint on Linux-right:

```

1 vsrx-right> show security flow session dynamic-application junos:SSH
2 Session ID: 1438, Policy name: tenant-1/10, Timeout: 1798, Session State: Valid
3 In: 10.1.2.10/34762 --> 10.2.1.10/22;tcp, L3VPN VRF Group: tenant-1, If: st0.0, Pkts: 18,
4     Bytes: 3612,
5 Out: 10.2.1.10/22 --> 10.1.2.10/34762;tcp, If: ge-0/0/2.82, Pkts: 22, Bytes: 4800,
6 Total sessions: 1

```

And corresponding vSRX-right session close log:

```

1 RT_FLOW - RT_FLOW_SESSION_CLOSE [junos@2636.1.1.1.2.129 reason="TCP FIN" source-address
2     ="10.1.2.10" source-port="34762" destination-address="10.2.1.10" destination-port="22"
3     connection-tag="0" service-name="junos-ssh" nat-source-address="10.1.2.10" nat-source-port
4     ="34762" nat-destination-address="10.2.1.10" nat-destination-port="22" nat-connection-tag
5     ="0" src-nat-rule-type="N/A" src-nat-rule-name="N/A" dst-nat-rule-type="N/A" dst-nat-rule-
6     name="N/A" protocol-id="6" policy-name="tenant-1" source-zone-name="infra" destination-zone-
7     name="tenant-2-trust" session-id="1438" packets-from-client="24" bytes-from-client="4056"
8     packets-from-server="27" bytes-from-server="5332" elapsed-time="118" application="SSH"
9     nested-application="UNKNOWN" username="N/A" roles="N/A" packet-incoming-interface="st0.0"

```

```
encrypted="No" application-category="Remote-Access" application-sub-category="Command"
application-risk="4" application-characteristics="Supports File Transfer;Known
Vulnerabilities;Capable of Tunneling;" secure-web-proxy-session-type="NA" peer-session-id
="0" peer-source-address="0.0.0.0" peer-source-port="0" peer-destination-address="0.0.0.0"
peer-destination-port="0" hostname="NA NA" src-vrf-grp="tenant-1" dst-vrf-grp="N/A" tunnel-
inspection="Off" tunnel-inspection-policy-set="root" session-flag="0" source-tenant="N/A"
destination-service="N/A" user-type="N/A"]
```

Things get more interesting when vSRX-left tenant-1 is connecting to tenant-2 also bound to vSRX-left. Traffic is heading towards vSRX-right for policy enforcement and inter-VRF communication and U-turns back to vSRX-left.

Note: without IPSEC, and vSRX-left being, for example, a Juniper QFX device, this would have been a firewall between VRFs in a DC scenario.

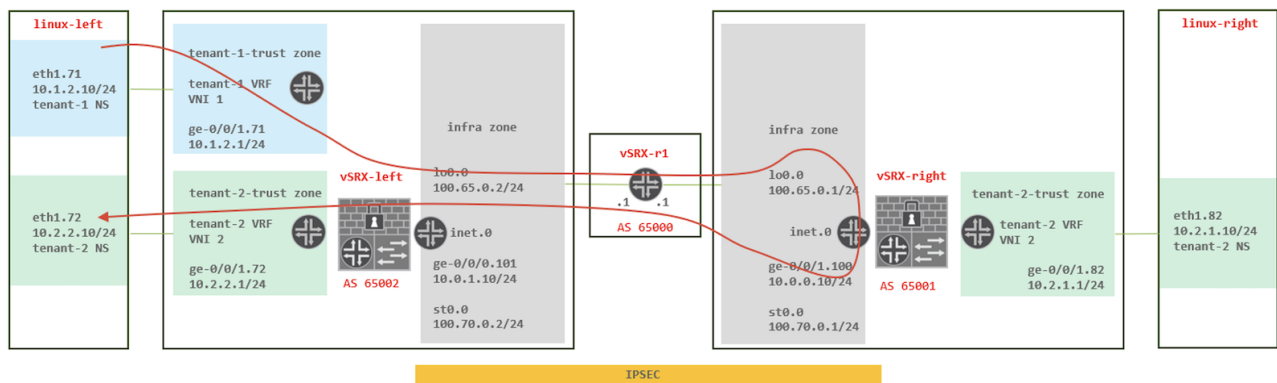


Figure 5: U-turn on vSRX-right blue - green tenant VRF

Policy on vSRX-right is evaluated in the infra to infra intra-zone context (another option would have been global policy with an option to narrow down to zones using from-zone to-zone matching). Application Identification based SSH, HTTP, and ICMP ping are permitted.

```
1 [edit security policies from-zone infra to-zone infra]
2 vsrx-right# show | display set relative
3 set policy tenant-1 apply-groups any-permit-log
4 set policy tenant-1 match dynamic-application junos:SSH
5 set policy tenant-1 match dynamic-application junos:HTTP
6 set policy tenant-1 match dynamic-application junos:ICMP-ECHO
7 set policy tenant-1 match source-l3vpn-vrf-group tenant-1
8 set policy tenant-1 match destination-l3vpn-vrf-group tenant-1
9 set policy tenant-1 match destination-l3vpn-vrf-group tenant-2
10 set policy tenant-1 match destination-l3vpn-vrf-group tenant-3
```

Session listing on vSRX-right shows the effective U-turn of the traffic on the st0.0 IPSEC underlay tunnel adapter and tenant-1 / tenant-2 VRFs. SRX handles that as a single session across routing instances:

```
1 vsrx-right> show security flow session dynamic-application junos:SSH
2 Session ID: 1441, Policy name: tenant-1/7, Timeout: 1796, Session State: Valid
3 In: 10.1.2.10/55526 --> 10.2.2.10/22;tcp, L3VPN VRF Group: tenant-1, If: st0.0, Pkts: 18,
   Bytes: 3612,
4 Out: 10.2.2.10/22 --> 10.1.2.10/55526;tcp, L3VPN VRF Group: tenant-2, If: st0.0, Pkts: 17,
   Bytes: 4388,
```

Implication on vSRX-left is tracking the same session twice due to different ingress interface/zone.

```
1 vsrx-left> show security flow session destination-port 22
2 Session ID: 1454, Policy name: tenant-1/4, Timeout: 1782, Session State: Valid
3 In: 10.1.2.10/55526 --> 10.2.2.10/22;tcp, If: ge-0/0/1.71, Pkts: 18, Bytes: 3612,
4 Out: 10.2.2.10/22 --> 10.1.2.10/55526;tcp, L3VPN VRF Group: tenant-1, If: st0.0, Pkts: 17,
   Bytes: 4388,
5
```

```

6 Session ID: 1455, Policy name: tenant-2/7, Timeout: 1782, Session State: Valid
7 In: 10.1.2.10/55526 --> 10.2.2.10/22;tcp, L3VPN VRF Group: tenant-2, If: st0.0, Pkts: 18,
  Bytes: 3612,
8 Out: 10.2.2.10/22 --> 10.1.2.10/55526;tcp, If: ge-0/0/1.72, Pkts: 17, Bytes: 4388,
9 Total sessions: 2

```

Two related vSRX-left session close logs:

```

1 RT_FLOW_SESSION_CLOSE [junos@2636.1.1.1.2.129 reason="TCP FIN" source-address="10.1.2.10" source
  -port="55526" destination-address="10.2.2.10" destination-port="22" connection-tag="0"
  service-name="junos-ssh" nat-source-address="10.1.2.10" nat-source-port="55526" nat-
  destination-address="10.2.2.10" nat-destination-port="22" nat-connection-tag="0" src-nat-
  rule-type="N/A" src-nat-rule-name="N/A" dst-nat-rule-type="N/A" dst-nat-rule-name="N/A"
  protocol-id="6" policy-name="tenant-1" source-zone-name="tenant-1-trust" destination-zone-
  name="infra" session-id="1454" packets-from-client="28" bytes-from-client="4372" packets-
  from-server="28" bytes-from-server="5376" elapsed-time="477" application="UNKNOWN" nested-
  application="UNKNOWN" username="N/A" roles="N/A" packet-incoming-interface="ge-0/0/1.71"
  encrypted="UNKNOWN" application-category="N/A" application-sub-category="N/A" application-
  risk="-1" application-characteristics="N/A" secure-web-proxy-session-type="NA" peer-session-
  id="0" peer-source-address="0.0.0.0" peer-source-port="0" peer-destination-address="0.0.0.0"
  peer-destination-port="0" hostname="NA NA" src-vrf-grp="N/A" dst-vrf-grp="tenant-1" tunnel-
  inspection="Off" tunnel-inspection-policy-set="root" session-flag="0" source-tenant="N/A"
  destination-service="N/A" user-type="N/A"]
2
3 RT_FLOW - RT_FLOW_SESSION_CLOSE [junos@2636.1.1.1.2.129 reason="TCP FIN" source-address
  ="10.1.2.10" source-port="55526" destination-address="10.2.2.10" destination-port="22"
  connection-tag="0" service-name="junos-ssh" nat-source-address="10.1.2.10" nat-source-port
  ="55526" nat-destination-address="10.2.2.10" nat-destination-port="22" nat-connection-tag
  ="0" src-nat-rule-type="N/A" src-nat-rule-name="N/A" dst-nat-rule-type="N/A" dst-nat-rule-
  name="N/A" protocol-id="6" policy-name="tenant-2" source-zone-name="infra" destination-zone-
  name="tenant-2-trust" session-id="1455" packets-from-client="28" bytes-from-client="4372"
  packets-from-server="28" bytes-from-server="5376" elapsed-time="477" application="UNKNOWN"
  nested-application="UNKNOWN" username="N/A" roles="N/A" packet-incoming-interface="st0.0"
  encrypted="UNKNOWN" application-category="N/A" application-sub-category="N/A" application-
  risk="-1" application-characteristics="N/A" secure-web-proxy-session-type="NA" peer-session-
  id="0" peer-source-address="0.0.0.0" peer-source-port="0" peer-destination-address="0.0.0.0"
  peer-destination-port="0" hostname="NA NA" src-vrf-grp="tenant-2" dst-vrf-grp="N/A" tunnel-
  inspection="Off" tunnel-inspection-policy-set="root" session-flag="0" source-tenant="N/A"
  destination-service="N/A" user-type="N/A"]

```

1.6 Tenant-1 (blue) to Tenant-3 (red) – Overlapping IP Prefixes

Enhancement of the previous example is cross-tenant communication when IP prefixes overlap - specifically, tenant-1 and tenant-2 communication:

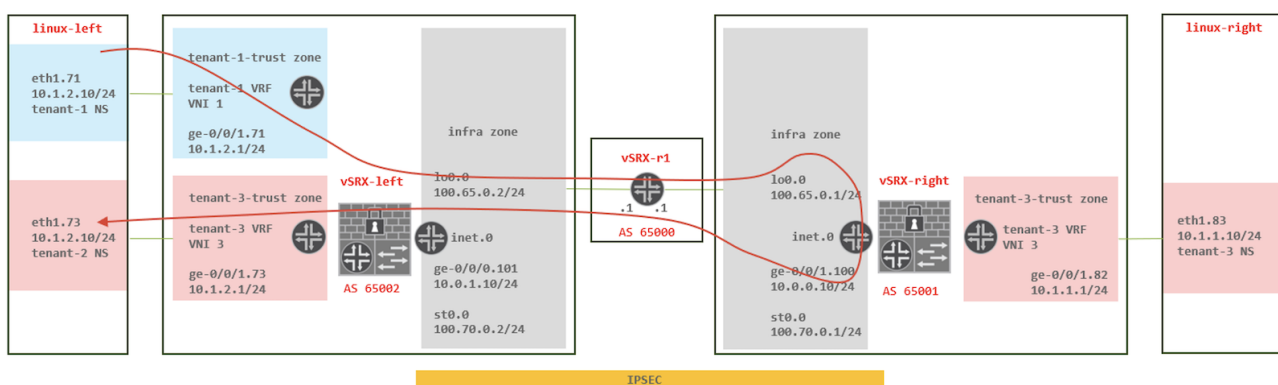


Figure 6: Overlapping prefix handling on vSRX-right for blue/red tenant VRFs

The solution for tenant-1 connecting to tenant-2 is to leverage the bi-directional nature of static NAT, where tenant-1 10.1.0.0/16 is represented by the arbitrary prefix 10.21/16 and correspondingly tenant-3 10.1.0.0/16 by the 10.23/16 prefix



Figure 7: Details of vSRX-right NAT approach

vSRX-right static NAT settings (complete config in Appendix 3. also covers traffic from local interface):

```
1 [edit security nat static]
2 vsrx-right# show | display set relative
3 set rule-set tenant-1-vrf from routing-group tenant-1
4 set rule-set tenant-1-vrf rule tenant-1-vrf match source-address 10.1.0.0/16
5 set rule-set tenant-1-vrf rule tenant-1-vrf match destination-address 10.23.0.0/16
6 set rule-set tenant-1-vrf rule tenant-1-vrf then static-nat prefix 10.1.0.0/16
7 set rule-set tenant-1-vrf rule tenant-1-vrf then static-nat prefix routing-instance tenant-3
8
9 set rule-set tenant-3-vrf from routing-group tenant-3
10 set rule-set tenant-3-vrf rule tenant-3-vrf match source-address 10.1.0.0/16
11 set rule-set tenant-3-vrf rule tenant-3-vrf match destination-address 10.21.0.0/16
12 set rule-set tenant-3-vrf rule tenant-3-vrf then static-nat prefix 10.1.0.0/16
13 set rule-set tenant-3-vrf rule tenant-3-vrf then static-nat prefix routing-instance tenant-1
```

The same security policy as in the previous example in the infra to infra zone context, matching on tenant-3 VRF:

```
1 [edit security policies from-zone infra to-zone infra]
2 vsrx-right# show | display set relative
3 set policy tenant-1 apply-groups any-permit-log
4 set policy tenant-1 match dynamic-application junos:SSH
5 set policy tenant-1 match dynamic-application junos:HTTP
6 set policy tenant-1 match dynamic-application junos:ICMP-ECHO
7 set policy tenant-1 match source-l3vpn-vrf-group tenant-1
8 set policy tenant-1 match destination-l3vpn-vrf-group tenant-1
9 set policy tenant-1 match destination-l3vpn-vrf-group tenant-2
10 set policy tenant-1 match destination-l3vpn-vrf-group tenant-3
```

Sample session listing on vSRX-right with static NAT in action:

```
1 srx-right> show security flow session dynamic-application junos:SSH
2 Session ID: 127, Policy name: tenant-1/7, Timeout: 1706, Session State: Valid
3 In: 10.1.2.10/50440 --> 10.23.2.10/22;tcp, L3VPN VRF Group: tenant-1, If: st0.0, Pkts: 19,
   Bytes: 3672,
4 Out: 10.1.2.10/22 --> 10.21.2.10/50440;tcp, L3VPN VRF Group: tenant-3, If: st0.0, Pkts: 18,
   Bytes: 4476,
5 Total sessions: 10
```

Session listing from vSRX-left where naturally two sessions are seen as in the previous example:

```
1 vsrx-left> show security flow session destination-port 22
2 Session ID: 128, Policy name: tenant-1/4, Timeout: 1718, Session State: Valid
3 In: 10.1.2.10/50440 --> 10.23.2.10/22;tcp, If: ge-0/0/1.71, Pkts: 19, Bytes: 3672,
4 Out: 10.23.2.10/22 --> 10.1.2.10/50440;tcp, L3VPN VRF Group: tenant-1, If: ge-0/0/0.101, Pkts:
   18, Bytes: 4476,
5
6 Session ID: 129, Policy name: tenant-3/9, Timeout: 1718, Session State: Valid
7 In: 10.21.2.10/50440 --> 10.1.2.10/22;tcp, L3VPN VRF Group: tenant-3, If: ge-0/0/0.101, Pkts:
   19, Bytes: 3672,
8 Out: 10.1.2.10/22 --> 10.21.2.10/50440;tcp, If: ge-0/0/1.73, Pkts: 18, Bytes: 4476,
9 Total sessions: 2
```

And related close-log from vSRX-right:

```

1 RT_FLOW_SESSION_CLOSE [junos@2636.1.1.1.2.129 reason="TCP FIN" source-address="10.1.2.10" source-
  -port="50440" destination-address="10.23.2.10" destination-port="22" connection-tag="0"
  service-name="junos-ssh" nat-source-address="10.21.2.10" nat-source-port="50440" nat-
  destination-address="10.1.2.10" nat-destination-port="22" nat-connection-tag="0" src-nat-
  rule-type="static rule" src-nat-rule-name="tenant-3-vrf" dst-nat-rule-type="static rule" dst-
  -nat-rule-name="tenant-1-vrf" protocol-id="6" policy-name="tenant-1" source-zone-name="infra
  " destination-zone-name="infra" session-id="127" packets-from-client="27" bytes-from-client
  ="4292" packets-from-server="27" bytes-from-server="5300" elapsed-time="243" application="
  SSH" nested-application="UNKNOWN" username="N/A" roles="N/A" packet-incoming-interface="st0
  .0" encrypted="No" application-category="Remote-Access" application-sub-category="Command"
  application-risk="4" application-characteristics="Supports File Transfer;Known
  Vulnerabilities;Capable of Tunneling;" secure-web-proxy-session-type="NA" peer-session-id
  ="0" peer-source-address="0.0.0.0" peer-source-port="0" peer-destination-address="0.0.0.0"
  peer-destination-port="0" hostname="NA NA" src-vrf-grp="tenant-1" dst-vrf-grp="tenant-3"
  tunnel-inspection="Off" tunnel-inspection-policy-set="root" session-flag="0" source-tenant="
  N/A" destination-service="N/A" user-type="N/A"]

```

1.7 MTU Considerations

Due to the VXLAN header and IPSEC header + trailer overhead, the effective MTU for endpoint traffic is reduced. This is not a concern in a DC scenario where large MTUs must be used. There are a couple of ways to address this in scenarios where a large MTU can't be configured.

This is the header size breakdown for an MTU of 1500 bytes in the specific setup:

Header Type	Size (B)
Ether	14
802.1q	4
IPv4	20
ESP	54
UDP	8
VXLAN	8
Ether	14
IPv4	20
TCP header	32
TCP payload	1344

TCP can be easily influenced by adjusting the TCP Maximum Segment Size (MSS). In a scenario with VXLAN and IPSEC AES256-GCM transform, the exact TCP-MSS value is 1356B (effectively on the wire 1344B due to an extra 12B in the TCP header above the minimum of 20B). For clear-text without the IPSEC header, MSS would have been 1410B. These settings can be configured on the SRX either globally or per policy.

Global MSS settings for IPSEC scenario:

```

1 set security flow tcp-mss ipsec-vpn mss 1356

```

Global MSS settings for clear-text scenario (not applicable to DC with appropriate MTU!):

```

1 set security flow tcp-mss all-tcp mss 1410

```

Without MSS or for non-TCP traffic, Path MTU discovery by host OS is next. Oversized Don't Fragment flagged IP packets get ICMP need frag needed message:

```

1  14:16:56.765270 52:54:00:18:ce:48 > 52:54:00:c2:78:8f, ethertype 802.1Q (0x8100), length 1606:
   vlan 81, p 0, ethertype IPv4 (0x0800), (tos 0x0, ttl 64, id 56951, offset 0, flags [DF],
   proto TCP (6), length 1588)
2  10.1.1.10.53000 > 10.1.2.10.22: Flags [P.], seq 41:1577, ack 41, win 502, options [nop,nop,TS
   val 4180791614 ecr 4000098031], length 1536
3
4  14:16:56.765395 52:54:00:c2:78:8f > 52:54:00:18:ce:48, ethertype 802.1Q (0x8100), length 74:
   vlan 81, p 0, ethertype IPv4 (0x0800), (tos 0x0, ttl 254, id 14449, offset 0, flags [none],
   proto ICMP (1), length 56)
5  10.1.1.1 > 10.1.1.10: ICMP 10.1.2.10 unreachable - need to frag (mtu 1396), length 36
6  (tos 0x0, ttl 64, id 56951, offset 0, flags [DF], proto TCP (6), length 1500)
7  10.1.1.10.53000 > 10.1.2.10.22: [|tcp]

```

And IP stack caches the PMTU information:

```

1  linux-right:~# t1 ip rou get 10.1.2.10
2  10.1.2.10 via 10.1.1.1 dev eth1.81 src 10.1.1.10 uid 0
3  cache expires 273sec mtu 1396

```

It should be noted that ICMP type 3 code 4 replies are rate-limited on SRX firewalls due to the potential for abuse as a DoS attack vector. Relying on this method for production is NOT recommended. Using MSS clamping, as described above, is the most efficient mechanism. Alternatively, using flow fragmentation and reassembly is another method, which is described next.

In this specific example setup, the maximum UDP payload that can be passed via a 1500B MTU is 1368B. Pre-fragmentation occurs when crossing this boundary. For instance, if 2B oversized UDP packets are sent from Linux-right to Linux-left using the hping tool and captured on Linux-left using tcpdump:

```

1  linux-right:~# t1 hping3 -2 -s 1024 --keep 10.1.2.10 -d 1370 -p 1024
2  HPING 10.1.2.10 (eth1.81 10.1.2.10): udp mode set, 28 headers + 1370 data bytes
3
4  linux-left:~# tcpdump -n -i eth1 not arp -v
5  tcpdump: listening on eth1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
6  16:34:22.042015 IP (tos 0x0, ttl 63, id 38195, offset 0, flags [+], proto UDP (17), length 1396)
7  10.1.1.10.1024 > 10.1.2.10.1024: UDP, length 1370
8  16:34:22.042017 IP (tos 0x0, ttl 63, id 38195, offset 1376, flags [none], proto UDP (17), length
   22)
9  10.1.1.10 > 10.1.2.10: ip-proto-17

```

If fragmentation is not desirable, SRX can reassemble the packets towards egress:

```

1  set security flow force-ip-reassembly

```

Then unlike capture above, receiving side gets reassembled traffic. However, please note that there are performance implications of using both pre-fragmentation and reassembly.

```

1  linux-left:~# tcpdump -n -i eth1 not arp -v
2  tcpdump: listening on eth1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
3  16:40:35.099305 IP (tos 0x0, ttl 63, id 49943, offset 0, flags [none], proto UDP (17), length
   1398)
4  10.1.1.10.1024 > 10.1.2.10.1024: UDP, length 1370
5  16:40:36.099391 IP (tos 0x0, ttl 63, id 15129, offset 0, flags [none], proto UDP (17), length
   1398)
6  10.1.1.10.1024 > 10.1.2.10.1024: UDP, length 1370

```

1.8 Appendix 1 – SRX Transit VXLAN Tunnel Inspection

Let's have a brief look at the transit EVPN VXLAN tunnel inspection feature. For this purpose, IPSEC underlay is disabled, and vSRX-r1 is performing the inspections of VXLAN encapsulated traffic between vSRX-left and vSRX-right loopback interfaces. The scenario looked at is vSRX-left blue tenant-1 talking to green tenant-2:

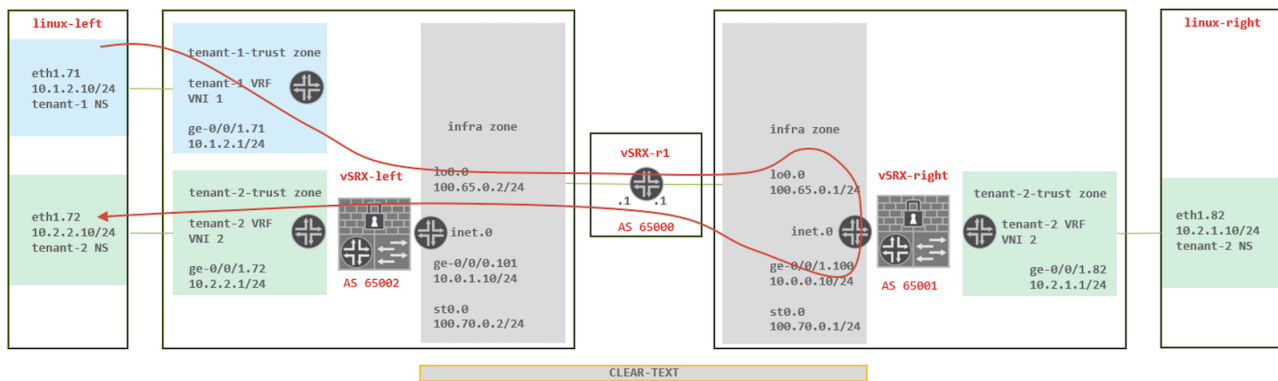


Figure 8: vSRX-r1 inspection and enforcement of unencrypted VXLAN

The security policy on vSRX-right from infra to infra zone looks as follows, permitting, besides others, also HTTP. In our arbitrary scenario, HTTP is not a concern if the transit is encrypted; however, it becomes problematic when passing unencrypted towards the WAN:

```

1 [edit security policies from-zone infra to-zone infra]
2 vsrx-right# show | display set relative
3 set policy tenant-1 apply-groups any-permit-log
4 set policy tenant-1 match dynamic-application junos:SSH
5 set policy tenant-1 match dynamic-application junos:HTTP
6 set policy tenant-1 match dynamic-application junos:ICMP-ECHO
7 set policy tenant-1 match source-l3vpn-vrf-group tenant-1
8 set policy tenant-1 match destination-l3vpn-vrf-group tenant-1
9 set policy tenant-1 match destination-l3vpn-vrf-group tenant-2
10 set policy tenant-1 match destination-l3vpn-vrf-group tenant-3

```

To address this, vSRX-r1 prohibits VXLAN encapsulated HTTP communication and permits the rest. Global router like any/any/permit policy pointing to the VXLAN tunnel inspection profile:

```

1 [edit security policies global]
2 vsrx-r1# show | display set
3 set security policies global policy all match source-address any
4 set security policies global policy all match destination-address any
5 set security policies global policy all match application any
6 set security policies global policy all then permit tunnel-inspection insp-prof-1

```

Inspection profile matching on any VNI and related policy-set – effectively rule-base for inspected transit VXLAN traffic:

```

1 [edit security tunnel-inspection]
2 vsrx-r1# show | display set relative
3 set inspection-profile insp-prof-1 vxlan pol-set-1 policy-set pol-set-1
4 set inspection-profile insp-prof-1 vxlan pol-set-1 vni any
5
6 [edit security policies policy-set pol-set-1]
7 vsrx-r1# show | display set relative
8 set policy reject-some match source-address any
9 set policy reject-some match destination-address any
10 set policy reject-some match application any
11 set policy reject-some match dynamic-application junos:HTTP
12 set policy reject-some then reject
13 set policy reject-some then log session-init
14 set policy permit-all match source-address any
15 set policy permit-all match destination-address any
16 set policy permit-all match application any
17 set policy permit-all match dynamic-application any
18 set policy permit-all then permit
19 set policy permit-all then log session-close

```

vSRX-r1 discarding the HTTP traffic:

```

1 vsrx-r1> monitor security packet-drop
2 Starting packet drop:

```

```
3 10:56:18.761458:LSYS-ID-00 10.2.2.10/80-->10.1.2.10/46284;tcp,ipid-38722,ge-0/0/0.101,Dropped by
POLICY:discarded by dynapp policy reject-some
```

Related log of rejected HTTP communication:

```
1 RT_FLOW_SESSION_DENY [junos@2636.1.1.1.2.129 source-address="10.1.2.10" source-port="46284"
destination-address="10.2.2.10" destination-port="80" connection-tag="fcd" service-name="
junos-http" protocol-id="6" icmp-type="0" policy-name="reject-some" source-zone-name="all"
destination-zone-name="all" application="HTTP" nested-application="UNKNOWN" username="N/A"
roles="N/A" packet-incoming-interface="ge-0/0/0.101" encrypted="No" reason="Rejected by
policy" session-id="409" application-category="Web" application-sub-category="N/A"
application-risk="4" application-characteristics="Can Leak Information;Supports File
Transfer;Prone to Misuse;Known Vulnerabilities;Carrier of Malware;Capable of Tunneling;" src
-vrf-grp="N/A" dst-vrf-grp="N/A" source-tenant="N/A" destination-service="N/A" user-type="N/
A"]
```

For completeness, example session listing of permitted SSH session (matching permit-all policy) between tenant-1 left and tenant-2 right:

```
1 vsrx-r1> show security flow session tunnel-inspection-type vxlan
2 Total sessions: 0
3
4 root@vsrx-r1> show security flow session tunnel-inspection-type vxlan
5 Session ID: 392, Policy name: permit-all/6, Timeout: 1796, Session State: Valid
6 In: 10.1.2.10/55412 --> 10.2.2.10/22;tcp, Conn Tag: 0xfcd, If: ge-0/0/0.101, Pkts: 36, Bytes:
7 7224,
8 Type: VXLAN, VNI: 2, Tunnel Session ID: 393
9 Out: 10.2.2.10/22 --> 10.1.2.10/55412;tcp, Conn Tag: 0xfcd, If: ge-0/0/1.100, Pkts: 32, Bytes:
10 8672,
11 Type: VXLAN, VNI: 1, Tunnel Session ID: 395
```

1.9 Appendix 2 – Linux Network Namespace Sample Configs

For convenience, execution of commands in Linux network namespaces is done using shell aliases folding complete command line into two characters (t1, t2, t3) with a parameter of command to be executed. For example:

```
1 linux-right:~# cat .bashrc
2 alias t1='ip netns exec tenant-1'
3 alias t2='ip netns exec tenant-2'
4 alias t3='ip netns exec tenant-3'
5
6 linux-right:~# alias
7 alias t1='ip netns exec tenant-1'
8 alias t2='ip netns exec tenant-2'
9 alias t3='ip netns exec tenant-3'
10
11 linux-right:~# t1 ping 10.1.2.10
12 PING 10.1.2.10 (10.1.2.10) 56(84) bytes of data:
13 64 bytes from 10.1.2.10: icmp_seq=1 ttl=63 time=2.24 ms
```

Namespaces can be configured for example as part of interface settings (on Debian Linux typically in /etc/network/interfaces). For testing purposes there are also SSH, nginx and inetd daemons spawn in namespaces, otherwise namespace doesn't have any bound service by default. Self-explanatory example from the demo setup:

```
1 linux-right:~# cat /etc/network/interfaces
2 source /etc/network/interfaces.d/*
3
4 auto lo
5 iface lo inet loopback
6
7 auto eth0
8 iface eth0 inet static
9     address 172.30.100.208/24
10     dns-nameservers 172.30.100.105
11     gateway 172.30.100.1
12
```

```

13 auto eth1
14 iface eth1 inet manual
15
16 #tenant-1
17     up ip netns add tenant-1
18     up ip link add link eth1 name eth1.81 type vlan id 81
19     up ip link set eth1.81 netns tenant-1
20     up ip netns exec tenant-1 ip link set eth1.81 up
21     up ip netns exec tenant-1 ip addr add 10.1.1.10/24 dev eth1.81
22     up ip netns exec tenant-1 ip rou add default via 10.1.1.1
23     up $( sleep 3; ip netns exec tenant-1 /usr/sbin/sshd -o PidFile=/run/sshd-tenant-1.pid ) &
24     up $( sleep 3; ip netns exec tenant-1 /usr/sbin/nginx ) &
25     up $( sleep 3; ip netns exec tenant-1 /usr/sbin/inetd ) &
26
27 #tenant-2
28     up ip netns add tenant-2
29     up ip link add link eth1 name eth1.82 type vlan id 82
30     up ip link set eth1.82 netns tenant-2
31     up ip netns exec tenant-2 ip link set eth1.82 up
32     up ip netns exec tenant-2 ip addr add 10.2.1.10/24 dev eth1.82
33     up ip netns exec tenant-2 ip rou add default via 10.2.1.1
34     up $( sleep 3; ip netns exec tenant-2 /usr/sbin/sshd -o PidFile=/run/sshd-tenant-2.pid ) &
35     up $( sleep 3; ip netns exec tenant-2 /usr/sbin/nginx ) &
36     up $( sleep 3; ip netns exec tenant-2 /usr/sbin/inetd ) &
37
38 #tenant-3
39     up ip netns add tenant-3
40     up ip link add link eth1 name eth1.83 type vlan id 83
41     up ip link set eth1.83 netns tenant-3
42     up ip netns exec tenant-3 ip link set eth1.83 up
43     up ip netns exec tenant-3 ip addr add 10.1.1.10/24 dev eth1.83
44     up ip netns exec tenant-3 ip rou add default via 10.1.1.1
45     up $( sleep 3; ip netns exec tenant-3 /usr/sbin/sshd -o PidFile=/run/sshd-tenant-3.pid ) &
46     up $( sleep 3; ip netns exec tenant-3 /usr/sbin/nginx ) &
47     up $( sleep 3; ip netns exec tenant-3 /usr/sbin/inetd ) &

```

1.10 Appendix 3 – SRX Configuration Files

Please note that configuration files are pristine, demo/testbed grade, real life requires tuning, hardening, and appropriate qualification. vSRX-left has tenants configured in a form of Junos groups and takes different sample approach to tighten BGP signaling than vSRX-right.

vSRX-left:

```

1 set version 23.2R2.21
2 set groups tenant-1 security policies from-zone tenant-1-trust to-zone infra apply-groups any-permit
3 set groups tenant-1 security policies from-zone tenant-1-trust to-zone infra policy tenant-1 match destination-l3vpn-vrf-group tenant-1
4 set groups tenant-1 security policies from-zone infra to-zone tenant-1-trust apply-groups any-permit
5 set groups tenant-1 security policies from-zone infra to-zone tenant-1-trust policy tenant-1 match source-l3vpn-vrf-group tenant-1
6 set groups tenant-1 security zones security-zone tenant-1-trust tcp-rst
7 set groups tenant-1 security zones security-zone tenant-1-trust interfaces ge-0/0/1.71 host-inbound-traffic system-services ping
8 set groups tenant-1 security l3vpn vrf-group tenant-1 vrf tenant-1
9 set groups tenant-1 interfaces ge-0/0/1 vlan-tagging
10 set groups tenant-1 interfaces ge-0/0/1 unit 71 description tenant-1
11 set groups tenant-1 interfaces ge-0/0/1 unit 71 vlan-id 71
12 set groups tenant-1 interfaces ge-0/0/1 unit 71 family inet address 10.1.2.1/24
13 set groups tenant-1 policy-options policy-statement export-tenant-1 term 1 from interface ge-0/0/1.71
14 set groups tenant-1 policy-options policy-statement export-tenant-1 term 1 then accept
15 set groups tenant-1 policy-options policy-statement export-tenant-1 term 100 then reject
16 set groups tenant-1 routing-instances tenant-1 instance-type vrf
17 set groups tenant-1 routing-instances tenant-1 protocols evpn ip-prefix-routes advertise direct-nextthop
18 set groups tenant-1 routing-instances tenant-1 protocols evpn ip-prefix-routes encapsulation vxlan
19 set groups tenant-1 routing-instances tenant-1 protocols evpn ip-prefix-routes vni 1
20 set groups tenant-1 routing-instances tenant-1 protocols evpn ip-prefix-routes export export-tenant-1
21 set groups tenant-1 routing-instances tenant-1 interface ge-0/0/1.71

```

```

22 set groups tenant-1 routing-instances tenant-1 route-distinguisher 100.65.0.1:1
23 set groups tenant-1 routing-instances tenant-1 vrf-target target:65001:1
24 set groups tenant-2 security policies from-zone tenant-2-trust to-zone infra apply-groups any-
  permit
25 set groups tenant-2 security policies from-zone tenant-2-trust to-zone infra policy tenant-2
  match destination-l3vpn-vrf-group tenant-2
26 set groups tenant-2 security policies from-zone infra to-zone tenant-2-trust apply-groups any-
  permit
27 set groups tenant-2 security policies from-zone infra to-zone tenant-2-trust policy tenant-2
  match source-l3vpn-vrf-group tenant-2
28 set groups tenant-2 security zones security-zone tenant-2-trust tcp-rst
29 set groups tenant-2 security zones security-zone tenant-2-trust interfaces ge-0/0/1.72 host-
  inbound-traffic system-services ping
30 set groups tenant-2 security l3vpn vrf-group tenant-2 vrf tenant-2
31 set groups tenant-2 interfaces ge-0/0/1 unit 72 description tenant-2
32 set groups tenant-2 interfaces ge-0/0/1 unit 72 vlan-id 72
33 set groups tenant-2 interfaces ge-0/0/1 unit 72 family inet address 10.2.2.1/24
34 set groups tenant-2 policy-options policy-statement export-tenant-2 term 1 from interface ge
  -0/0/1.72
35 set groups tenant-2 policy-options policy-statement export-tenant-2 term 1 then accept
36 set groups tenant-2 policy-options policy-statement export-tenant-2 term 100 then reject
37 set groups tenant-2 routing-instances tenant-2 instance-type vrf
38 set groups tenant-2 routing-instances tenant-2 protocols evpn ip-prefix-routes advertise direct-
  nexthop
39 set groups tenant-2 routing-instances tenant-2 protocols evpn ip-prefix-routes encapsulation
  vxlan
40 set groups tenant-2 routing-instances tenant-2 protocols evpn ip-prefix-routes vni 2
41 set groups tenant-2 routing-instances tenant-2 protocols evpn ip-prefix-routes export export-
  tenant-2
42 set groups tenant-2 routing-instances tenant-2 interface ge-0/0/1.72
43 set groups tenant-2 routing-instances tenant-2 route-distinguisher 100.65.0.1:2
44 set groups tenant-2 routing-instances tenant-2 vrf-target target:65001:2
45 set groups tenant-3 security policies from-zone tenant-3-trust to-zone infra apply-groups any-
  permit
46 set groups tenant-3 security policies from-zone tenant-3-trust to-zone infra policy tenant-3
  match destination-l3vpn-vrf-group tenant-3
47 set groups tenant-3 security policies from-zone infra to-zone tenant-3-trust apply-groups any-
  permit
48 set groups tenant-3 security policies from-zone infra to-zone tenant-3-trust policy tenant-3
  match source-l3vpn-vrf-group tenant-3
49 set groups tenant-3 security zones security-zone tenant-3-trust tcp-rst
50 set groups tenant-3 security zones security-zone tenant-3-trust interfaces ge-0/0/1.73 host-
  inbound-traffic system-services ping
51 set groups tenant-3 security l3vpn vrf-group tenant-3 vrf tenant-3
52 set groups tenant-3 interfaces ge-0/0/1 unit 73 description tenant-3
53 set groups tenant-3 interfaces ge-0/0/1 unit 73 vlan-id 73
54 set groups tenant-3 interfaces ge-0/0/1 unit 73 family inet address 10.1.2.1/24
55 set groups tenant-3 policy-options policy-statement export-tenant-3 term 1 from interface ge
  -0/0/1.73
56 set groups tenant-3 policy-options policy-statement export-tenant-3 term 1 then accept
57 set groups tenant-3 policy-options policy-statement export-tenant-3 term 100 then reject
58 set groups tenant-3 routing-instances tenant-3 instance-type vrf
59 set groups tenant-3 routing-instances tenant-3 protocols evpn ip-prefix-routes advertise direct-
  nexthop
60 set groups tenant-3 routing-instances tenant-3 protocols evpn ip-prefix-routes encapsulation
  vxlan
61 set groups tenant-3 routing-instances tenant-3 protocols evpn ip-prefix-routes vni 3
62 set groups tenant-3 routing-instances tenant-3 protocols evpn ip-prefix-routes export export-
  tenant-3
63 set groups tenant-3 routing-instances tenant-3 interface ge-0/0/1.73
64 set groups tenant-3 routing-instances tenant-3 route-distinguisher 100.65.0.1:3
65 set groups tenant-3 routing-instances tenant-3 vrf-target target:65001:3
66 set groups any-permit security policies from-zone <*> to-zone <*> policy <*> match source-
  address any
67 set groups any-permit security policies from-zone <*> to-zone <*> policy <*> match destination-
  address any
68 set groups any-permit security policies from-zone <*> to-zone <*> policy <*> match application
  any
69 set groups any-permit security policies from-zone <*> to-zone <*> policy <*> then permit
70 set groups any-permit security policies from-zone <*> to-zone <*> policy <*> then log session-
  close
71 set apply-groups tenant-1
72 set apply-groups tenant-2
73 set apply-groups tenant-3
74 set system host-name vsrx-left
75 set system services ssh root-login allow
76 set system services ssh sftp-server
77 set system services ssh client-alive-interval 120
78 set system time-zone Europe/Amsterdam
79 set system name-server 172.30.100.105
80 set system syslog file messages any any

```

```

81 set system syslog file messages archive size 5m
82 set system syslog file messages archive files 5
83 set system syslog file traffic any any
84 set system syslog file traffic match RT_
85 set system syslog file traffic archive size 1m
86 set system syslog file traffic archive files 4
87 set system syslog file traffic structured-data brief
88 set system ntp server 172.30.100.105
89 set services application-identification
90 set security ike proposal ike-prop authentication-method pre-shared-keys
91 set security ike proposal ike-prop dh-group group20
92 set security ike proposal ike-prop encryption-algorithm aes-256-gcm
93 set security ike proposal ike-prop lifetime-seconds 28800
94 set security ike policy ike-policy proposals ike-prop
95 set security ike policy ike-policy pre-shared-key ascii-text ""
96 set security ike gateway vsrx-right ike-policy ike-policy
97 set security ike gateway vsrx-right address 10.0.0.10
98 set security ike gateway vsrx-right dead-peer-detection probe-idle-tunnel
99 set security ike gateway vsrx-right external-interface ge-0/0/0.101
100 set security ike gateway vsrx-right version v2-only
101 set security ipsec proposal ipsec-prop encryption-algorithm aes-256-gcm
102 set security ipsec proposal ipsec-prop lifetime-seconds 3600
103 set security ipsec policy ipsec-policy perfect-forward-secrecy keys group20
104 set security ipsec policy ipsec-policy proposals ipsec-prop
105 set security ipsec vpn vsrx-right bind-interface st0.0
106 set security ipsec vpn vsrx-right ike gateway vsrx-right
107 set security ipsec vpn vsrx-right ike ipsec-policy ipsec-policy
108 set security ipsec vpn vsrx-right establish-tunnels immediately
109 set security alg dns disable
110 set security alg ftp disable
111 set security alg h323 disable
112 set security alg mgcp disable
113 set security alg msrpc disable
114 set security alg sunrpc disable
115 set security alg rtsp disable
116 set security alg sccp disable
117 set security alg sip disable
118 set security alg talk disable
119 set security alg tftp disable
120 set security alg pptp disable
121 set security flow tcp-mss ipsec-vpn mss 1356
122 set security flow tcp-session strict-syn-check
123 set security forwarding-process enhanced-services-mode
124 set security address-book global address vSRX-r1 10.0.1.1/32
125 set security address-book global address vSRX1-lo0 100.65.0.1/32
126 set security address-book global address vSRX1-st0 100.70.0.1/32
127 set security address-book global address-set EVPN address vSRX-r1
128 set security address-book global address-set EVPN address vSRX1-lo0
129 set security address-book global address-set EVPN address vSRX1-st0
130 set security policies global policy infra match source-address EVPN
131 set security policies global policy infra match destination-address any
132 set security policies global policy infra match application junos-bgp
133 set security policies global policy infra match application bfd-mhop
134 set security policies global policy infra match from-zone infra
135 set security policies global policy infra match to-zone infra
136 set security policies global policy infra then permit
137 set security policies global policy infra-ping match source-address any
138 set security policies global policy infra-ping match destination-address any
139 set security policies global policy infra-ping match application junos-icmp-ping
140 set security policies global policy infra-ping then permit
141 set security policies global policy drop-log match source-address any
142 set security policies global policy drop-log match destination-address any
143 set security policies global policy drop-log match application any
144 set security policies global policy drop-log then deny
145 set security policies global policy drop-log then log session-init
146 set security zones security-zone infra host-inbound-traffic system-services ping
147 set security zones security-zone infra host-inbound-traffic system-services ike
148 set security zones security-zone infra host-inbound-traffic protocols bgp
149 set security zones security-zone infra host-inbound-traffic protocols bfd
150 set security zones security-zone infra interfaces ge-0/0/0.101
151 set security zones security-zone infra interfaces lo0.0
152 set security zones security-zone infra interfaces st0.0
153 set interfaces ge-0/0/0 description r1
154 set interfaces ge-0/0/0 vlan-tagging
155 set interfaces ge-0/0/0 unit 101 description underlay-phy
156 set interfaces ge-0/0/0 unit 101 vlan-id 101
157 set interfaces ge-0/0/0 unit 101 family inet address 10.0.1.10/24
158 set interfaces fxp0 unit 0 family inet address 172.30.100.215/24
159 set interfaces lo0 unit 0 description overlay
160 set interfaces lo0 unit 0 family inet address 100.65.0.2/32
161 set interfaces st0 unit 0 description underlay-ipsec

```

```

162 set interfaces st0 unit 0 family inet address 100.70.0.2/24
163 set policy-options policy-statement export-underlay term 1 from interface lo0.0
164 set policy-options policy-statement export-underlay term 1 then accept
165 set policy-options policy-statement export-underlay term 100 then reject
166 set applications application bfd-mhop term 1 protocol udp
167 set applications application bfd-mhop term 1 destination-port 4784
168 set protocols bgp group underlay bfd-liveness-detection minimum-interval 500
169 set protocols bgp group underlay bfd-liveness-detection multiplier 3
170 set protocols bgp group underlay neighbor 10.0.1.1 description r1
171 set protocols bgp group underlay neighbor 10.0.1.1 export export-underlay
172 set protocols bgp group underlay neighbor 10.0.1.1 peer-as 65000
173 set protocols bgp group underlay neighbor 10.0.1.1 local-as 65002
174 set protocols bgp group underlay neighbor 100.70.0.1 export export-underlay
175 set protocols bgp group underlay neighbor 100.70.0.1 description vsrx-right-st0
176 set protocols bgp group underlay neighbor 100.70.0.1 peer-as 65001
177 set protocols bgp group underlay neighbor 100.70.0.1 local-as 65002
178 set protocols bgp group overlay multihop
179 set protocols bgp group overlay local-address 100.65.0.2
180 set protocols bgp group overlay family evpn signaling
181 set protocols bgp group overlay bfd-liveness-detection minimum-interval 500
182 set protocols bgp group overlay bfd-liveness-detection multiplier 5
183 set protocols bgp group overlay neighbor 100.65.0.1 peer-as 65101
184 set protocols bgp group overlay neighbor 100.65.0.1 local-as 65102
185 set routing-options static route 0.0.0.0/0 next-hop 172.30.100.1

```

vSRX-right:

```

1 set version 23.2R2.21
2 set groups any-permit-log security policies from-zone <*> to-zone <*> policy <*> match source-
  address any
3 set groups any-permit-log security policies from-zone <*> to-zone <*> policy <*> match
  destination-address any
4 set groups any-permit-log security policies from-zone <*> to-zone <*> policy <*> match
  application any
5 set groups any-permit-log security policies from-zone <*> to-zone <*> policy <*> then permit
6 set groups any-permit-log security policies from-zone <*> to-zone <*> policy <*> then log
  session-close
7 set system host-name vsrx-right
8 set system services ssh root-login allow
9 set system services ssh sftp-server
10 set system services ssh client-alive-interval 120
11 set system time-zone Europe/Amsterdam
12 set system name-server 172.30.100.105
13 set system syslog file messages any any
14 set system syslog file messages archive size 5m
15 set system syslog file messages archive files 5
16 set system syslog file traffic any any
17 set system syslog file traffic match RT_
18 set system syslog file traffic archive size 1m
19 set system syslog file traffic archive files 4
20 set system syslog file traffic structured-data brief
21 set system ntp server 172.30.100.105
22 set security log mode stream-event
23 set security log report
24 set security ike proposal ike-prop authentication-method pre-shared-keys
25 set security ike proposal ike-prop dh-group group20
26 set security ike proposal ike-prop encryption-algorithm aes-256-gcm
27 set security ike proposal ike-prop lifetime-seconds 28800
28 set security ike policy ike-policy proposals ike-prop
29 set security ike policy ike-policy pre-shared-key ascii-text ""
30 set security ike gateway vsrx-left ike-policy ike-policy
31 set security ike gateway vsrx-left address 10.0.1.10
32 set security ike gateway vsrx-left dead-peer-detection probe-idle-tunnel
33 set security ike gateway vsrx-left external-interface ge-0/0/1.100
34 set security ike gateway vsrx-left version v2-only
35 set security ipsec proposal ipsec-prop encryption-algorithm aes-256-gcm
36 set security ipsec proposal ipsec-prop lifetime-seconds 3600
37 set security ipsec policy ipsec-policy perfect-forward-secrecy keys group20
38 set security ipsec policy ipsec-policy proposals ipsec-prop
39 set security ipsec vpn vsrx-left bind-interface st0.0
40 set security ipsec vpn vsrx-left ike gateway vsrx-left
41 set security ipsec vpn vsrx-left ike ipsec-policy ipsec-policy
42 set security address-book global address n_tenant-1 10.1.0.0/16
43 set security address-book global address n_tenant-3 10.2.0.0/16
44 set security address-book global address n_tenant-2 10.2.0.0/16
45 set security alg dns disable
46 set security alg ftp disable
47 set security alg h323 disable
48 set security alg mgcp disable
49 set security alg msrpc disable

```

```

50 set security alg sunrpc disable
51 set security alg rtsp disable
52 set security alg sccp disable
53 set security alg sip disable
54 set security alg talk disable
55 set security alg tftp disable
56 set security alg pptp disable
57 set security flow tcp-mss ipsec-vpn mss 1356
58 set security flow tcp-session strict-syn-check
59 set security nat static rule-set tenant-1-vrf from routing-group tenant-1
60 set security nat static rule-set tenant-1-vrf rule tenant-1-vrf match source-address 10.1.0.0/16
61 set security nat static rule-set tenant-1-vrf rule tenant-1-vrf match destination-address
   10.23.0.0/16
62 set security nat static rule-set tenant-1-vrf rule tenant-1-vrf then static-nat prefix
   10.1.0.0/16
63 set security nat static rule-set tenant-1-vrf rule tenant-1-vrf then static-nat prefix routing-
   instance tenant-3
64 set security nat static rule-set tenant-3-vrf from routing-group tenant-3
65 set security nat static rule-set tenant-3-vrf rule tenant-3-vrf match source-address 10.1.0.0/16
66 set security nat static rule-set tenant-3-vrf rule tenant-3-vrf match destination-address
   10.21.0.0/16
67 set security nat static rule-set tenant-3-vrf rule tenant-3-vrf then static-nat prefix
   10.1.0.0/16
68 set security nat static rule-set tenant-3-vrf rule tenant-3-vrf then static-nat prefix routing-
   instance tenant-1
69 set security nat static rule-set tenant-1-loc from zone tenant-1-trust
70 set security nat static rule-set tenant-1-loc rule tenant-1-loc match source-address 10.1.0.0/16
71 set security nat static rule-set tenant-1-loc rule tenant-1-loc match destination-address
   10.23.0.0/16
72 set security nat static rule-set tenant-1-loc rule tenant-1-loc then static-nat prefix
   10.1.0.0/16
73 set security nat static rule-set tenant-1-loc rule tenant-1-loc then static-nat prefix routing-
   instance tenant-3
74 set security nat static rule-set tenant-3-loc from zone tenant-3-trust
75 set security nat static rule-set tenant-3-loc rule tenant-3-loc match source-address 10.1.0.0/16
76 set security nat static rule-set tenant-3-loc rule tenant-3-loc match destination-address
   10.21.0.0/16
77 set security nat static rule-set tenant-3-loc rule tenant-3-loc then static-nat prefix
   10.1.0.0/16
78 set security nat static rule-set tenant-3-loc rule tenant-3-loc then static-nat prefix routing-
   instance tenant-1
79 set security forwarding-process enhanced-services-mode
80 set security policies from-zone tenant-1-trust to-zone infra policy tenant-1-any apply-groups
   any-permit-log
81 set security policies from-zone tenant-1-trust to-zone infra policy tenant-1-any match
   destination-l3vpn-vrf-group tenant-1
82 set security policies from-zone tenant-1-trust to-zone infra policy tenant-1-limited apply-
   groups any-permit-log
83 set security policies from-zone tenant-1-trust to-zone infra policy tenant-1-limited match
   dynamic-application junos:SSH
84 set security policies from-zone tenant-1-trust to-zone infra policy tenant-1-limited match
   dynamic-application junos:ICMP-ECHO
85 set security policies from-zone tenant-1-trust to-zone infra policy tenant-1-limited match
   destination-l3vpn-vrf-group tenant-2
86 set security policies from-zone tenant-1-trust to-zone infra policy tenant-1-limited match
   destination-l3vpn-vrf-group tenant-3
87 set security policies from-zone infra to-zone tenant-1-trust policy tenant-1 apply-groups any-
   permit-log
88 set security policies from-zone infra to-zone tenant-1-trust policy tenant-1 match source-l3vpn-
   vrf-group tenant-1
89 set security policies from-zone infra to-zone infra policy tenant-1 apply-groups any-permit-log
90 set security policies from-zone infra to-zone infra policy tenant-1 match dynamic-application
   junos:SSH
91 set security policies from-zone infra to-zone infra policy tenant-1 match dynamic-application
   junos:HTTP
92 set security policies from-zone infra to-zone infra policy tenant-1 match dynamic-application
   junos:ICMP-ECHO
93 set security policies from-zone infra to-zone infra policy tenant-1 match source-l3vpn-vrf-group
   tenant-1
94 set security policies from-zone infra to-zone infra policy tenant-1 match destination-l3vpn-vrf-
   group tenant-1
95 set security policies from-zone infra to-zone infra policy tenant-1 match destination-l3vpn-vrf-
   group tenant-2
96 set security policies from-zone infra to-zone infra policy tenant-1 match destination-l3vpn-vrf-
   group tenant-3
97 set security policies from-zone tenant-2-trust to-zone infra policy tenant-2 apply-groups any-
   permit-log
98 set security policies from-zone tenant-2-trust to-zone infra policy tenant-2 match destination-
   l3vpn-vrf-group tenant-2
99 set security policies from-zone infra to-zone tenant-2-trust policy tenant-2 apply-groups any-
   permit-log

```

```

100 set security policies from-zone infra to-zone tenant-2-trust policy tenant-2 match source-l3vpn-
    vrf-group tenant-2
101 set security policies from-zone infra to-zone tenant-2-trust policy tenant-1 apply-groups any-
    permit-log
102 set security policies from-zone infra to-zone tenant-2-trust policy tenant-1 match dynamic-
    application junos:SSH
103 set security policies from-zone infra to-zone tenant-2-trust policy tenant-1 match dynamic-
    application junos:ICMP-ECHO
104 set security policies from-zone infra to-zone tenant-2-trust policy tenant-1 match source-l3vpn-
    vrf-group tenant-1
105 set security policies from-zone tenant-3-trust to-zone infra policy tenant-3 apply-groups any-
    permit-log
106 set security policies from-zone tenant-3-trust to-zone infra policy tenant-3 match destination-
    l3vpn-vrf-group tenant-3
107 set security policies from-zone infra to-zone tenant-3-trust policy tenant-3 apply-groups any-
    permit-log
108 set security policies from-zone infra to-zone tenant-3-trust policy tenant-3 match source-l3vpn-
    vrf-group tenant-3
109 set security policies from-zone infra to-zone tenant-3-trust policy tenant-1 apply-groups any-
    permit-log
110 set security policies from-zone infra to-zone tenant-3-trust policy tenant-1 match dynamic-
    application junos:SSH
111 set security policies from-zone infra to-zone tenant-3-trust policy tenant-1 match dynamic-
    application junos:ICMP-ECHO
112 set security policies from-zone infra to-zone tenant-3-trust policy tenant-1 match source-l3vpn-
    vrf-group tenant-1
113 set security policies from-zone tenant-1-trust to-zone tenant-2-trust policy tenant-1 apply-
    groups any-permit-log
114 set security policies from-zone tenant-1-trust to-zone tenant-2-trust policy tenant-1 match
    dynamic-application junos:SSH
115 set security policies from-zone tenant-1-trust to-zone tenant-2-trust policy tenant-1 match
    dynamic-application junos:ICMP-ECHO
116 set security policies from-zone tenant-1-trust to-zone tenant-3-trust policy tenant-1 apply-
    groups any-permit-log
117 set security policies from-zone tenant-1-trust to-zone tenant-3-trust policy tenant-1 match
    dynamic-application junos:SSH
118 set security policies from-zone tenant-1-trust to-zone tenant-3-trust policy tenant-1 match
    dynamic-application junos:ICMP-ECHO
119 set security policies global policy cleanup-src-vrfs match source-address any
120 set security policies global policy cleanup-src-vrfs match destination-address any
121 set security policies global policy cleanup-src-vrfs match application any
122 set security policies global policy cleanup-src-vrfs match dynamic-application any
123 set security policies global policy cleanup-src-vrfs match source-l3vpn-vrf-group tenant-1
124 set security policies global policy cleanup-src-vrfs match source-l3vpn-vrf-group tenant-2
125 set security policies global policy cleanup-src-vrfs match source-l3vpn-vrf-group tenant-3
126 set security policies global policy cleanup-src-vrfs then reject
127 set security policies global policy cleanup-src-vrfs then log session-init
128 set security policies global policy cleanup-dst-vrfs match source-address any
129 set security policies global policy cleanup-dst-vrfs match destination-address any
130 set security policies global policy cleanup-dst-vrfs match application any
131 set security policies global policy cleanup-dst-vrfs match dynamic-application any
132 set security policies global policy cleanup-dst-vrfs match destination-l3vpn-vrf-group tenant-1
133 set security policies global policy cleanup-dst-vrfs match destination-l3vpn-vrf-group tenant-2
134 set security policies global policy cleanup-dst-vrfs match destination-l3vpn-vrf-group tenant-3
135 set security policies global policy cleanup-dst-vrfs then reject
136 set security policies global policy cleanup-dst-vrfs then log session-init
137 set security policies global policy infra description "control plane permit"
138 set security policies global policy infra match source-address any
139 set security policies global policy infra match destination-address any
140 set security policies global policy infra match application junos-bgp
141 set security policies global policy infra match application junos-icmp-ping
142 set security policies global policy infra match application bfd-mhop
143 set security policies global policy infra match dynamic-application any
144 set security policies global policy infra match from-zone infra
145 set security policies global policy infra match to-zone infra
146 set security policies global policy infra then permit
147 set security policies global policy infra then log session-init
148 set security policies global policy infra then log session-close
149 set security policies global policy infra-reject-log description "final touch"
150 set security policies global policy infra-reject-log match source-address any
151 set security policies global policy infra-reject-log match destination-address any
152 set security policies global policy infra-reject-log match application any
153 set security policies global policy infra-reject-log match dynamic-application any
154 set security policies global policy infra-reject-log then reject
155 set security policies global policy infra-reject-log then log session-init
156 set security policies pre-id-default-policy then log session-close
157 set security zones security-zone infra host-inbound-traffic system-services ping
158 set security zones security-zone infra host-inbound-traffic system-services ike
159 set security zones security-zone infra host-inbound-traffic protocols bgp
160 set security zones security-zone infra host-inbound-traffic protocols bfd
161 set security zones security-zone infra interfaces lo0.0

```

```

162 set security zones security-zone infra interfaces ge-0/0/1.100
163 set security zones security-zone infra interfaces st0.0
164 set security zones security-zone infra enable-reverse-reroute
165 set security zones security-zone tenant-1-trust tcp-rst
166 set security zones security-zone tenant-1-trust interfaces ge-0/0/2.81 host-inbound-traffic
    system-services ping
167 set security zones security-zone tenant-2-trust tcp-rst
168 set security zones security-zone tenant-2-trust interfaces ge-0/0/2.82 host-inbound-traffic
    system-services ping
169 set security zones security-zone tenant-3-trust tcp-rst
170 set security zones security-zone tenant-3-trust interfaces ge-0/0/2.83 host-inbound-traffic
    system-services ping
171 set security l3vpn vrf-group tenant-1 vrf tenant-1
172 set security l3vpn vrf-group tenant-2 vrf tenant-2
173 set security l3vpn vrf-group tenant-3 vrf tenant-3
174 set interfaces ge-0/0/1 vlan-tagging
175 set interfaces ge-0/0/1 unit 100 description underlay-phy
176 set interfaces ge-0/0/1 unit 100 vlan-id 100
177 set interfaces ge-0/0/1 unit 100 family inet address 10.0.0.10/24
178 set interfaces ge-0/0/2 vlan-tagging
179 set interfaces ge-0/0/2 unit 81 description tenant-1
180 set interfaces ge-0/0/2 unit 81 vlan-id 81
181 set interfaces ge-0/0/2 unit 81 family inet address 10.1.1.1/24
182 set interfaces ge-0/0/2 unit 82 description tenant-2
183 set interfaces ge-0/0/2 unit 82 vlan-id 82
184 set interfaces ge-0/0/2 unit 82 family inet address 10.2.1.1/24
185 set interfaces ge-0/0/2 unit 83 description tenant-3
186 set interfaces ge-0/0/2 unit 83 vlan-id 83
187 set interfaces ge-0/0/2 unit 83 family inet address 10.1.1.1/24
188 set interfaces fxp0 unit 0 family inet address 172.30.193.193/24
189 set interfaces lo0 unit 0 description overlay
190 set interfaces lo0 unit 0 family inet address 100.65.0.1/32
191 set interfaces st0 unit 0 description underlay-ipsec
192 set interfaces st0 unit 0 multipoint
193 set interfaces st0 unit 0 family inet address 100.70.0.1/24
194 set policy-options policy-statement export-tenant-1 term 1 from route-filter 0.0.0.0/0 exact
195 set policy-options policy-statement export-tenant-1 term 1 then accept
196 set policy-options policy-statement export-tenant-1 term 100 then reject
197 set policy-options policy-statement export-tenant-2 term 1 from route-filter 0.0.0.0/0 exact
198 set policy-options policy-statement export-tenant-2 term 1 then accept
199 set policy-options policy-statement export-tenant-2 term 100 then reject
200 set policy-options policy-statement export-tenant-3 term 1 from route-filter 0.0.0.0/0 exact
201 set policy-options policy-statement export-tenant-3 term 1 then accept
202 set policy-options policy-statement export-tenant-3 term 100 then reject
203 set policy-options policy-statement export-underlay term 1 from route-filter 100.65.0.1/32 exact
204 set policy-options policy-statement export-underlay term 1 then accept
205 set policy-options policy-statement export-underlay term 100 then reject
206 set policy-options policy-statement reject-defaults term 1 from route-filter 0.0.0.0/0 exact
207 set policy-options policy-statement reject-defaults term 1 then reject
208 set policy-options policy-statement reject-defaults term 100 then accept
209 set routing-instances common-1 instance-type virtual-router
210 set routing-instances tenant-1 instance-type vrf
211 set routing-instances tenant-1 routing-options rib tenant-1.inet.0 static route 0.0.0.0/0 next-
    table common-1.inet.0
212 set routing-instances tenant-1 routing-options auto-export family inet unicast rib-group to-
    common-1
213 set routing-instances tenant-1 protocols evpn ip-prefix-routes advertise direct-nexthop
214 set routing-instances tenant-1 protocols evpn ip-prefix-routes encapsulation vxlan
215 set routing-instances tenant-1 protocols evpn ip-prefix-routes vni 1
216 set routing-instances tenant-1 protocols evpn ip-prefix-routes export export-tenant-1
217 set routing-instances tenant-1 interface ge-0/0/2.81
218 set routing-instances tenant-1 route-distinguisher 100.65.0.1:1
219 set routing-instances tenant-1 vrf-target target:65001:1
220 set routing-instances tenant-2 instance-type vrf
221 set routing-instances tenant-2 routing-options rib tenant-2.inet.0 static route 0.0.0.0/0 next-
    table common-1.inet.0
222 set routing-instances tenant-2 routing-options auto-export family inet unicast rib-group to-
    common-1
223 set routing-instances tenant-2 protocols evpn ip-prefix-routes advertise direct-nexthop
224 set routing-instances tenant-2 protocols evpn ip-prefix-routes encapsulation vxlan
225 set routing-instances tenant-2 protocols evpn ip-prefix-routes vni 2
226 set routing-instances tenant-2 protocols evpn ip-prefix-routes export export-tenant-2
227 set routing-instances tenant-2 interface ge-0/0/2.82
228 set routing-instances tenant-2 route-distinguisher 100.65.0.1:2
229 set routing-instances tenant-2 vrf-target target:65001:2
230 set routing-instances tenant-3 instance-type vrf
231 set routing-instances tenant-3 routing-options static route 0.0.0.0/0 discard
232 set routing-instances tenant-3 protocols evpn ip-prefix-routes advertise direct-nexthop
233 set routing-instances tenant-3 protocols evpn ip-prefix-routes encapsulation vxlan
234 set routing-instances tenant-3 protocols evpn ip-prefix-routes vni 3
235 set routing-instances tenant-3 protocols evpn ip-prefix-routes export export-tenant-3

```

```

236 set routing-instances tenant-3 interface ge-0/0/2.83
237 set routing-instances tenant-3 route-distinguisher 100.65.0.1:3
238 set routing-instances tenant-3 vrf-target target:65001:3
239 set applications application bfd-mhop term 1 protocol udp
240 set applications application bfd-mhop term 1 destination-port 4784
241 set protocols bgp group underlay bfd-liveness-detection minimum-interval 500
242 set protocols bgp group underlay bfd-liveness-detection multiplier 3
243 set protocols bgp group underlay neighbor 10.0.0.1 description r1
244 set protocols bgp group underlay neighbor 10.0.0.1 export export-underlay
245 set protocols bgp group underlay neighbor 10.0.0.1 peer-as 65000
246 set protocols bgp group underlay neighbor 10.0.0.1 local-as 65001
247 set protocols bgp group underlay neighbor 100.70.0.2 description vsrx-left-st0
248 set protocols bgp group underlay neighbor 100.70.0.2 export export-underlay
249 set protocols bgp group underlay neighbor 100.70.0.2 peer-as 65002
250 set protocols bgp group underlay neighbor 100.70.0.2 local-as 65001
251 set protocols bgp group overlay multihop
252 set protocols bgp group overlay local-address 100.65.0.1
253 set protocols bgp group overlay family evpn signaling
254 set protocols bgp group overlay bfd-liveness-detection minimum-interval 500
255 set protocols bgp group overlay bfd-liveness-detection multiplier 5
256 set protocols bgp group overlay neighbor 100.65.0.2 peer-as 65102
257 set protocols bgp group overlay neighbor 100.65.0.2 local-as 65101
258 set routing-options static route 0.0.0.0/0 next-hop 172.30.193.1
259 set routing-options rib-groups to-common-1 import-rib common-1.inet.0
260 set routing-options rib-groups to-common-1 import-policy reject-defaults

```

vSRX-r1

```

1 set version 23.2R2.21
2 set system host-name vsrx-r1
3 set system services ssh root-login allow
4 set system services ssh sftp-server
5 set system services ssh client-alive-interval 120
6 set system time-zone Europe/Amsterdam
7 set system name-server 172.30.100.105
8 set system syslog file messages any any
9 set system syslog file messages archive size 5m
10 set system syslog file messages archive files 5
11 set system syslog file traffic any any
12 set system syslog file traffic match RT_
13 set system syslog file traffic archive size 1m
14 set system syslog file traffic archive files 4
15 set system syslog file traffic structured-data brief
16 set system ntp server 172.30.100.105
17 set services application-identification
18 set security log mode stream-event
19 set security log report
20 set security alg dns disable
21 set security alg ftp disable
22 set security alg h323 disable
23 set security alg mgcp disable
24 set security alg msrpc disable
25 set security alg sunrpc disable
26 set security alg rtsp disable
27 set security alg sccp disable
28 set security alg sip disable
29 set security alg talk disable
30 set security alg tftp disable
31 set security alg pptp disable
32 set security flow tcp-session no-syn-check
33 set security flow tcp-session no-sequence-check
34 set security flow power-mode-disable
35 set security forwarding-process enhanced-services-mode
36 set security policies global policy all match source-address any
37 set security policies global policy all match destination-address any
38 set security policies global policy all match application any
39 set security policies global policy all then permit tunnel-inspection insp-prof-1
40 set security policies policy-set pol-set-1 policy reject-some match source-address any
41 set security policies policy-set pol-set-1 policy reject-some match destination-address any
42 set security policies policy-set pol-set-1 policy reject-some match application any
43 set security policies policy-set pol-set-1 policy reject-some match dynamic-application junos:
    HTTP
44 set security policies policy-set pol-set-1 policy reject-some then reject
45 set security policies policy-set pol-set-1 policy reject-some then log session-init
46 set security policies policy-set pol-set-1 policy permit-all match source-address any
47 set security policies policy-set pol-set-1 policy permit-all match destination-address any
48 set security policies policy-set pol-set-1 policy permit-all match application any
49 set security policies policy-set pol-set-1 policy permit-all match dynamic-application any
50 set security policies policy-set pol-set-1 policy permit-all then permit
51 set security policies policy-set pol-set-1 policy permit-all then log session-close

```

```

52 set security zones security-zone all host-inbound-traffic system-services ping
53 set security zones security-zone all host-inbound-traffic protocols bgp
54 set security zones security-zone all host-inbound-traffic protocols bfd
55 set security zones security-zone all interfaces ge-0/0/0.101
56 set security zones security-zone all interfaces ge-0/0/1.100
57 set security tunnel-inspection inspection-profile insp-prof-1 vxlan pol-set-1 policy-set pol-set
  -1
58 set security tunnel-inspection inspection-profile insp-prof-1 vxlan pol-set-1 vni any
59 set interfaces ge-0/0/0 vlan-tagging
60 set interfaces ge-0/0/0 unit 101 description left
61 set interfaces ge-0/0/0 unit 101 vlan-id 101
62 set interfaces ge-0/0/0 unit 101 family inet address 10.0.1.1/24
63 set interfaces ge-0/0/1 vlan-tagging
64 set interfaces ge-0/0/1 unit 100 description right
65 set interfaces ge-0/0/1 unit 100 vlan-id 100
66 set interfaces ge-0/0/1 unit 100 family inet address 10.0.0.1/24
67 set interfaces fxp0 unit 0 family inet address 172.30.100.192/24
68 set policy-options policy-statement export-hub term 1 from interface ge-0/0/0.101
69 set policy-options policy-statement export-hub term 1 then accept
70 set policy-options policy-statement export-hub term 2 from route-filter 100.65.0.0/16 orlonger
71 set policy-options policy-statement export-hub term 2 then accept
72 set policy-options policy-statement export-hub term 100 then reject
73 set policy-options policy-statement export-spokes term 1 from interface ge-0/0/1.100
74 set policy-options policy-statement export-spokes term 1 then accept
75 set policy-options policy-statement export-spokes term 2 from route-filter 100.65.0.0/16
  orlonger
76 set policy-options policy-statement export-spokes term 2 then accept
77 set policy-options policy-statement export-spokes term 100 then reject
78 set routing-instances vr instance-type virtual-router
79 set routing-instances vr protocols bgp group underlay bfd-liveness-detection minimum-interval
  500
80 set routing-instances vr protocols bgp group underlay bfd-liveness-detection multiplier 3
81 set routing-instances vr protocols bgp group underlay neighbor 10.0.1.10 export export-spokes
82 set routing-instances vr protocols bgp group underlay neighbor 10.0.1.10 peer-as 65002
83 set routing-instances vr protocols bgp group underlay neighbor 10.0.1.10 local-as 65000
84 set routing-instances vr protocols bgp group underlay neighbor 10.0.0.10 export export-hub
85 set routing-instances vr protocols bgp group underlay neighbor 10.0.0.10 peer-as 65001
86 set routing-instances vr protocols bgp group underlay neighbor 10.0.0.10 local-as 65000
87 set routing-instances vr interface ge-0/0/0.101
88 set routing-instances vr interface ge-0/0/1.100
89 set routing-options static route 0.0.0.0/0 next-hop 172.30.100.1

```

1.11 Useful links

- <https://www.juniper.net/documentation/us/en/software/junos/evpn-vxlan/topics/concept/evpn-route-type5-understanding.html>
- <https://www.juniper.net/documentation/us/en/software/junos/evpn-vxlan/topics/topic-map/evpn-vxlan-support-srx-series.html>
- <https://www.juniper.net/documentation/us/en/software/junos/nat/index.html>
- <https://www.debian.org/distrib/>
- <https://lwn.net/Articles/580893/>

1.12 Glossary

- BGP Border Gateway Protocol
- ESP Encapsulating Security Payload
- EVPN Ethernet VPN
- GRE Generic Routing Encapsulation
- HTTP Hyper Text Transfer Protocol
- ICMP Internet Control Message Protocol
- IKE Internet Key Exchange
- IPSEC IP security

- MPLS Multi-Protocol Layer Switching
- MSS Maximum Segment Size
- MTU Maximum Transmission Unit
- NAT Network Address Translation
- PMTU Path Maximum Transmission Unit
- SA Security Association
- SSH Secure Shell
- TCP Transmission Control Protocol
- UDP User Datagram Protocol
- VNI Virtual Network Identifier
- VRF Virtual Routing Forwarding
- VXLAN Virtual Extensible LAN
- WAN Wide Area Network

1.13 Acknowledgements

All the brilliant people who have been involved in the exploration of SRX EVPN/VXLAN Type 5 since the early days, namely: Mark Barrett, Ben Griffin, Damian Holdcroft, Pawel Kocimowski, Matthijs Nagel, Teddy Nicoghosian, Elisabeth Rodrigues, and Thorbjørn Zieger. A special shoutout goes to Mark and Matthijs for reviewing and providing valuable feedback. Finally, kudos to Juniper SRX engineering and product line management for pushing the envelope.



DC – AI – Routing – Switching – Security – Wireless – Automation

© Copyright 2026 Hewlett Packard Enterprise Development LP. The information contained herein is subject to change without notice. The only warranties for Hewlett Packard Enterprise products and services are set forth in the express warranty statements accompanying such products and services. Nothing herein should be construed as constituting an additional warranty.

Hewlett Packard Enterprise shall not be liable for technical or editorial errors or omissions contained herein.

Trademark acknowledgments, if needed. All third-party marks are the property of their respective owners.

HEWLETT PACKARD ENTERPRISE

HPE.com

HPE

